

ROLES OF AI IN ENHANCING CUSTOMER EXPERIENCE ON E-COMMERCE PLATFORMS IN COIMBATORE CITY

PROJECT 2025-2026

**Work done by
KEERTHANA R**

23BIS023

Under the Co-ordination of
Dr M Vaishnavi MBA., Ph.D.,

A Report Submitted in partial fulfillment of the requirements
for the award of the degree of
Bachelor of Business Administration — Information System



Department of Management Sciences

PSG COLLEGE OF ARTS & SCIENCE

An Autonomous College Affiliated to Bharathiar University

Accredited with A⁺⁺ Grade by NAAC (4th CYCLE)

College with Potential for Excellence (Status awarded by UGC)

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CERTIFICATE

This is to certify that project report entitled “**ROLES OF AI IN ENHANCING CUSTOMER EXPERIENCE ON E-COMMERCE PLATFORMS IN COIMBATORE CITY**” is a bonafide record of original research work done by **KEERTHANA R(23BIS023)** submitted in the partial fulfillment for the award of the degree of Bachelor of Business Administration - Information System

Faculty Coordinator

Head of the Department

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2. _____

DECLARATION

I **Ms. KEERTHANA R(23BIS023)** hereby declare that this Project entitled “**ROLES OF AI IN ENHANCING CUSTOMER EXPERIENCE ON E-COMMERCE PLATFORMS IN COIMBATORE CITY**” is an original work done by me under the co-ordination of Dr M Vaishnavi MBA., Ph.D., during my period of study 2025 - 2026.

Date:

Place:

Signature of the Student

ACKNOWLEDGEMENT

It is with great pleasure, deep satisfaction and gratitude I acknowledge the contributors towards the successful completion of the project

At this outlet, I wish to extend my heartfelt sincere thanks to **Dr. M SENGUTTUVAN.,** Principal, PSG College of Arts and Science and the members of the management for giving me the opportunity to do this project

It is a great honor to thank Head of the Department, **Dr. V. SRIPRIYA MBA, MPHIL, MSC (PSY), PHD,** Bachelor of Business Administration — Information System, PSG College of Arts & Science for her encouragement during the project.

I would like to place on record the sincere and earnest guidance extended to me, by my faculty co-ordinator **Dr M VAISHNAVI MBA., PH.D.,** Bachelor of Business Administration - Information System for his guidance and encouragement provided, which was a great support for the completion of this project.

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CHAPTER 01

1.1 INTRODUCTION TO THE STUDY

Based on the e-commerce sector, has grown exponentially in the last decade, fueled by technology developments and the growing use of online shopping. Amongst the developments, Artificial Intelligence (AI) has been a game-changer, transforming business-to-customer interactions and customer online shopping experiences. AI is the imitation of human intellectual processes by machines, most notably computer programs, in order to carry out tasks like learning, problem-solving, and decision-making. AI is being used in e-commerce to improve customer experience across various touch points by providing personalized, effective, and secure services.

AI contributes significantly to various aspects of customer experience improvement. Personalization is among the most evolved uses of AI, where programs scan customer information to suggest products according to preferences, browsing habits, and purchase history. This gives a highly customized shopping experience for individual consumers. Customer-centric technology, including chatbots and virtual assistants, helps immediately and cuts down response time, enhancing customer satisfaction.

KEY POINTS ABOUT AI (ARTIFICIAL INTELLIGENCE):

1.Types of AI

- Narrow AI (Weak AI): For particular tasks (e.g., virtual assistants such as Siri).
- General AI (Strong AI): Theoretical; able to carry out any intellectual task a human can perform.
- Super AI: A futuristic idea where AI will overtake human intelligence.

2.Applications of AI

- E-commerce: Personalized recommendations, Chabot's, fraud detection.
- Healthcare: Disease diagnosis, drug discovery, patient care.
- Finance: Fraud detection, automated trading, credit risk assessment.
- Transportation: Self-driving vehicles, traffic prediction.
- Manufacturing: Predictive maintenance, quality control.

3.AI Technologies

- Machine Learning: AI improves over time through learning from data.

- Natural Language Processing: Makes machines capable of understanding and processing human language (e.g., Chatbot's, translation).
- Computer Vision: Enables AI to understand images and videos (e.g., facial recognition).

4. Advantages of AI

- Improves efficiency and accuracy.
- Lowers human error.
- Offers 24/7 availability.
- Automates repetitive tasks.

5. Challenges and Risks

- Job displacement due to automation.
- Data privacy and security concerns.

OBJECTIVES OF AI (ARTIFICIAL INTELLIGENCE):

1. Automation of Repetitive Tasks
2. Enhance Decision-Making
3. Personalization and User Experience
4. Improve Efficiency and Productivity
5. Support Innovation and Problem-Solving
6. Strengthen Security and Fraud Detection
7. Natural Language Understanding and Communication
8. Predictive Analysis and Forecasting
9. Improve Accessibility

CUSTOMER EXPERIENCE ON E-COMMERCE PLATFORM:

Customer experience on e-commerce platform refers to the overall impression a customer forms based on their interactions with the online store throughout their shopping journey. This includes browsing products, ease of navigation, personalization, payment processes, delivery, and post-purchase support. A positive customer experience ensures customer satisfaction, builds loyalty, and increases repeat purchases.

E-COMMERCE PLATFORM

An e-commerce platform is an online solution that allows companies to operate and run online shops, where customers browse for products, make payments, and complete transactions online. Such platforms offer product catalog management, shopping cart, secure payment, order management, and customer support options. Examples of them include Shopify, Amazon, and Magento.

TYPES OF E-COMMERCE PLATFORMS:

1. Business-to-Consumer

- Straight selling of goods or services from companies to individual consumers.
- Examples: Amazon, Shopify, Flipkart

2. Business-to-Business

- Buying and selling between businesses, e.g., manufacturers to wholesalers or retailers.
- Examples: IndiaMART

3. Consumer-to-Consumer

- Individuals sell goods or services directly to fellow consumers.
- Examples: eBay, OLX, Facebook Marketplace

4. Consumer-to-Business

- Individuals sell goods or services to companies on a freelance or contractual basis.
- Examples: Upwork, Freelancer, Shutterstock

5. Business-to-Government

- Companies sell goods or services to government bodies through contracts and tenders.
- Examples: Government procurement platforms

6. Direct-to-Consumer

- Sell directly to consumers without intermediaries, typically on their own websites.
- Examples: Warby Parker, Glossier, Nike's online store

7. Marketplace Platforms

- Link multiple sellers to buyers in one platform, handling transactions and logistics.
- Examples: Amazon, Etsy, Walmart Marketplace

OBJECTIVES OF AN E-COMMERCE PLATFORM:

1. Online Transaction Facilitation
2. Customer Experience Improvement
3. Wider Market Coverage

- 4.Improved Operational Efficiency
- 5.Multi-Channel Support
- 6.Business Scalability
- 7.Data Aggregation and Analysis
- 8.Security and Trust Provision

ROLES OF AI IN ENHANCING CUSTOMER EXPERIENCE ON E-COMMERCE PLATFORMS:

1. Personalized Product Recommendations

AI scans customer behavior and recommends personalized products based on their interests, boosting conversion rates and satisfaction.

2. Dynamic Pricing

AI optimizes pricing strategies by scanning market trends, demand, and competitor prices in real-time and providing the best prices to customers.

3. Customer Support Automation

AI-driven Chabot's deliver real-time support, assisting customers in fixing problems easily and enhancing their experience.

4. Visual and Voice Search

AI allows customers to search for products via images or voice, quickening the process of search and making it more intuitive.

5. Inventory Management and Demand Forecasting

AI forecasts demand trends, enabling companies to manage stock effectively and prevent problems such as overstocking or stock outs.

1.2 NEED FOR THE STUDY

Because online shopping is becoming more and more important, customers are expecting more personalized experiences, and AI technologies like chatbots and machine learning have the potential to completely change the customer journey, research on how AI can improve the Customer experience in e-commerce is essential. Businesses can solve issues like algorithmic biases and privacy concerns while simultaneously increasing revenue, retention, and customer pleasure by having a better understanding of these responsibilities. The study analysis how artificial intelligence affects consumer's purchasing decisions and attract them towards the platform in unique way.

1.3 SCOPE OF THE STUDY

This research will seek to investigate the different applications of AI in improving customer experience on online retailing websites. It will explore the advantages and pitfalls of applying AI-based solutions, including ethical issues related to data privacy and algorithmic discrimination. The research will particularly address: Examining the effect of AI on many facets of the customer experience across product exploration, through checkout, to after-sales services. Assessing the success of AI-based solutions in enhancing customer satisfaction, loyalty, and sale. Analyzing the limitations and challenges of AI adoption in e-commerce, such as data needs, technical skills, and cost. Exploring the ethical aspects of AI usage in e-commerce, such as data privacy, algorithmic bias, and transparency.

1.4 OBJECTIVES OF THE STUDY

- To analyze how AI enhances customer support and service quality in e-commerce.
- To identify the key roles of AI in improving customer engagement on e-commerce platforms.
- To examine the impact of AI-driven personalization on customer experience.
- To assess the influence of AI-based safety and security features on customer trust and satisfaction.

1.5 LIMITATIONS OF THE STUDY

- The study is limited to a specific geographical area or sample, so results may not reflect the global e-commerce market.
- Rapid advancements in AI technology may make the findings less relevant over time.
- Resource, time, and sample size constraints may limit the scope and depth of the study.
- Ethical and privacy concerns may restrict open feedback from participants.

CHAPTER 02

REVIEW OF LITERATURE

1. Ibharim Halil (2023) studied how artificial intelligence (AI) in marketing affects people's decision to buy products online. The study showed that AI helps businesses understand customer behavior based on past data, which saves money and improves results. What made this study different is that it looked at how people think and feel when shopping on AI-powered websites — including their attitudes, values, and social influence. The results showed that AI can positively influence people's intention to make purchases online. This helps researchers, businesses, and marketers better understand how AI can shape online shopping behavior.

2. Yan Lee et al (2022) explored how artificial intelligence (AI) is transforming the e-commerce industry. The study looked at how AI is being used in online businesses and the impact it has made—both the benefits and the drawbacks. While AI has greatly boosted the growth of e-commerce, the research also pointed out that it's not a perfect solution. The authors emphasized the need to better understand the challenges and opportunities of AI to handle current issues and prepare for future ones in the e-commerce space.

3. Atul Nimbalkar (2022) studied how the use of artificial intelligence (AI) is becoming increasingly important in e-commerce. The research explains that AI, which focuses on creating smart machines that can think like humans, is used by online retailers for more than just product recommendations. It includes chatbots, virtual assistants, personalized services, review analysis, and inventory management. The study highlights that businesses are heavily investing in AI to stay competitive, as it plays a key role in improving customer experience and overall operations in e-commerce

4. Andreea Tamaduianu (2022) explored how artificial intelligence (AI) supports and enhances e-commerce. The study explains AI in detail, including its three development stages, and looks at how it's used in different areas like market research, customer service, inventory management, order handling, delivery, and attracting customers. It also highlights the challenges businesses face when trying to use AI effectively and discusses future possibilities for improving AI use in e-commerce.

5. Luísa Vitorino (2022) studied how artificial intelligence (AI) is shaping e-commerce fashion platforms. The research focused on how AI is impacting the shopping experience, especially for Gen-Z consumers, by influencing their decisions at every step. AI tools not only help brands manage supply and demand but also make the shopping experience more personalized and engaging—going beyond just choosing and buying products. The study highlights how tech innovation and wider access to AI are transforming both businesses and consumer behavior in fashion e-commerce.

6. Manasa R and Jayanthila Devi (2022) conducted a case study on how Amazon uses artificial Intelligence (AI) in retail. The study highlights that AI aims to help machines make smart decisions and perform tasks like problem-solving, understanding communication, and sensing the environment. Amazon uses around 30,000 robots in its warehouses, working independently through a mechanical autonomous system. The research found that AI benefits companies in three main ways: automating tasks, gaining insights from data, and improving interactions with customers and employees. This allows humans to focus more on complex, non-routine work.

7. Amber Srivastava (2021) studied how artificial intelligence (AI) is being used and its impact on the e-commerce sector, especially in India. Over the past decade, AI has transformed how online businesses operate—helping them manage large customer databases, use chatbots for interaction, and assist users in finding the right products. E-commerce companies are also using AI to improve recommendations, streamline sales, enhance search experiences, and detect fake reviews. The study offers insights into how AI is shaping online retail and influencing shopping platforms.

8. Gangzhi Guo (2021) researched how artificial intelligence (AI) is being used in live streaming e-commerce. With the rise of mobile internet and social media, live streaming has become a popular and fast-growing way to shop online. Whether on traditional platforms or short video apps, this trend is rapidly expanding. The study explains how AI provides the technical support needed for this growth and focuses on how AI is being applied specifically in live streaming e-commerce to enhance its development and effectiveness.

9. Gururaj P (2021) studied how artificial intelligence (AI) is being applied in e-commerce. The research highlights that while AI has existed for a long time, recent advancements are making work easier, faster, and more efficient. The study explores key AI applications in e-commerce, such as chatbots, smart logistics, recommendation systems, warehouse automation, visual search, and dynamic pricing. It provides an overview of the current state of e-commerce and analyzes how AI is being used today, along with its future potential in the industry.

10. Prateek Kalia (2021) in his book *Artificial Intelligence: Fundamentals and Applications*, explained how internet-based technologies, especially AI, have transformed the way businesses handle marketing, transactions, and customer service. In e-commerce, AI is widely used to perform tasks that were once done by humans. The book highlights AI's role in areas like market research, pricing, order processing, delivery, and customer support. It concludes that AI has significantly changed how e-commerce businesses operate, making processes more efficient and technology-driven.

11. Hedia Jegham and Ines Belgacem (2020) studied how artificial intelligence (AI) is transforming the entire e-commerce supply chain—from production and marketing to delivery and customer service. The research focused on the AI technologies behind tools like chatbots and recommendation systems, including neural networks and natural language processing. To show AI's real impact, the study analyzed four major e-commerce websites in the Middle East. The goal was to highlight how AI is making e-commerce platforms in the MENA region more efficient, attractive, and profitable.

12. Laith T. Khrais (2020) studied how artificial intelligence (AI) influences consumer demand in e-commerce, especially through personalized products and services. While AI has a strong impact, the research raises ethical concerns about how understandable or "explainable" these AI systems are. Using word cloud and concordance analysis, the study explored how scholars view the concept of explainability in AI. It emphasizes the need to improve machine learning models to make AI systems more transparent and interpretable, supporting the development of Explainable AI (XAI) in e-commerce.

13. Xia Song et al. (2019) studied how artificial intelligence (AI) is being used in e-commerce. With rapid advances in technology, AI is playing a growing role in daily life and business. The study highlights how AI has been successfully applied in e-commerce, bringing positive results. It focuses on key areas like AI assistants, smart logistics, recommendation systems, and dynamic pricing, showing how AI is improving the efficiency and operations of online businesses.

14. Dr. S. Vasantha (2019) studied the major advancements of artificial intelligence (AI) in India's e-commerce industry. As the market continues to grow rapidly, it's expected to become the world's second-largest by 2034. The study highlights how AI has helped strengthen global e-commerce, especially during tough economic times, by enabling accurate supply and demand forecasting. It also reviews the key AI developments in the Indian e-commerce sector during 2019, showing how past investments in AI research are now delivering real benefits.

15. Emily Johnson (2019) conducted a study to understand how artificial intelligence (AI) helps improve customer experience in online retail. Using data from 300 respondents and analysing it through descriptive statistics and regression, the study found that AI tools like product recommendations and virtual assistants significantly boost customer satisfaction and encourage repeat purchases in e-commerce.

CHAPTER 03

RESEARCH METHODOLOGY

This chapter gives detailed information about the methodology of the research study and it outlines the research design, population, and sample size, method of sampling technique and data collection and tools used for data collection.

RESEARCH DESIGN

For research design, descriptive research has been used as it describes how Artificial Intelligence (AI) enhances customer experiences on e-commerce platforms.

POPULATION

In this research the population is infinite because we cannot measure the number of individuals using e-commerce platforms and the population includes any member in Coimbatore city.

SAMPLING TECHNIQUE

The Convenience Sampling technique from Non -probability Sampling has been applied for this research because the sample (i.e people who use e-commerce platforms) can be easily drawn from the population.

SAMPLE SIZE

A sample size is 151 respondents were selected for the study. This number is considered statistically sufficient for analysis and reflects user experiences across various e-commerce platforms.

METHOD OF DATA COLLECTION

Both primary and secondary data are used for the study. The primary data are collected by using questionnaire method and secondary data are from articles, journals and publications.

TOOLS FOR DATA COLLECTION

Questionnaire has been used as a tool for data collection. The questionnaire contains structured questions related to the study.

TOOLS FOR ANALYSIS

Simple Percentage, Chi- Square test, Anova

CHI-SQUARE:

Chi-Square test is a statistical test used to check whether there is a significant relationship between two categorical variables. It compares observed frequencies (actual data) with expected frequencies (theoretical values).

FORMULA:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

SIMPLE PERCENTAGE:

Percentage analysis is a simple statistical method used to convert data into percentages for easy comparison. It helps in understanding proportions and making data more meaningful.

FORMULA:

$$\text{Percentage} = \frac{\text{Value}}{\text{Total Value}} \times 100$$

ANOVA:

ANOVA is a statistical technique used to compare the means of three or more groups and check if there is a significant difference among them.

FORMULA:

$$F = \frac{\text{Mean Square Between}}{\text{Mean Square Within}}$$

CHAPTER 04

DATA ANALYSIS & INTERPRETATION

The data is collected from 151 respondents in Coimbatore city. The collected data are arranged properly, analyzed systematically, and interpreted precisely. The research method utilized is simple and does not involve more comprehensive techniques.

TABLE 4.1: SHOWING THE AGE OF THE RESPONDENTS

S.NO	AGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	BELOW 18	14	9.3%
2	18-25	104	68.9%
3	26-35	29	19.2%
4	36-45	1	0.7%
5	ABOVE 45	3	2%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that the majority of respondents fall within the age group of 18 to 25 years, accounting for 68.9% of the total. A smaller portion, 19.2%, belong to the 26 to 35 age group, while 9.3% of the respondents are below 18 years of age. Only 0.7% fall in the 36 to 45 age group, and 2% are above 40 years.

CHART 4.1: SHOWING THE AGE OF THE RESPONDENTS

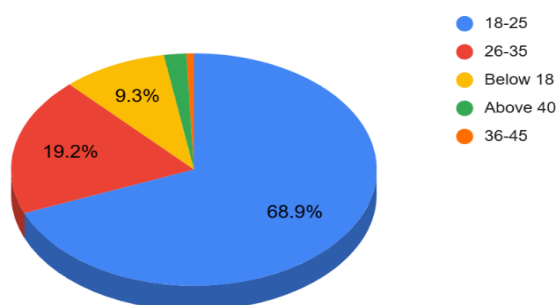


TABLE 4.2: SHOWING THE GENDER OF THE RESPONDENTS

S.NO	GENDER	RESPONDENTS	PERCENTAGE ANALYSIS
1	MALE	60	39.7%
2	FEMALE	91	60.3%
		151	100

Source: primary Data

INTERPRETATION

Table 4.2 indicates that 60.3% of respondents belong to female and 39.7% of respondents belong to male.

CHART 4.2: SHOWING THE GENDER OF THE RESPONDENTS

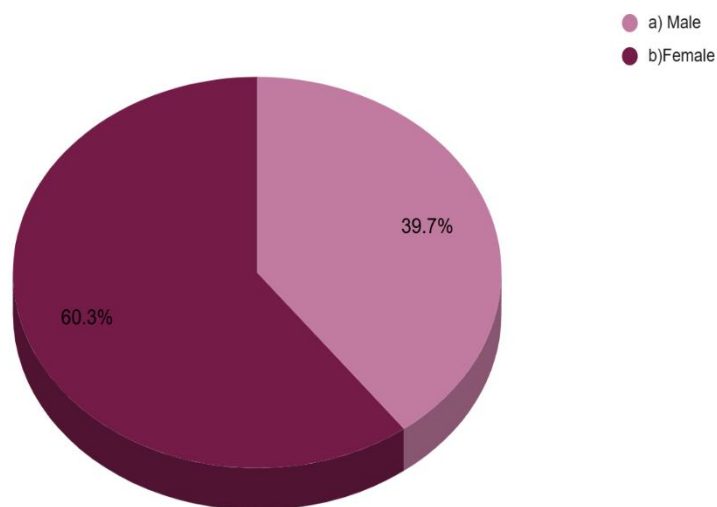


TABLE 4.3: SHOWING THE EDUCATIONAL QUALIFICATION OF THE RESPONDENTS

S.NO	EDUCATIONAL QUALIFICATION	RESPONDENTS	PERCENTAGE ANALYSIS
1	HIGH SCHOOL	32	21.2%
2	DIPLOMO	0	0%
3	UG	74	49.0%
4	PG	22	14.6%
5	OTHER	23	15.2%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that most of the people who took the survey are undergraduate students, making up 49% of the total. About 21.2% have studied up to high school, and 15.2% have other types of education. Around 14.6% are postgraduate students. There were no respondents with a diploma.

CHART 4.3: SHOWING THE EDUCATIONAL QUALIFICATION OF THE RESPONDENTS.

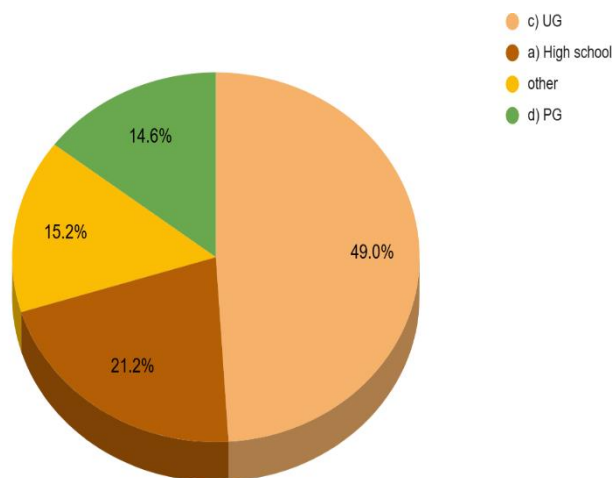


TABLE 4.4: SHOWING THE OCCUPATION OF THE RESPONDENTS

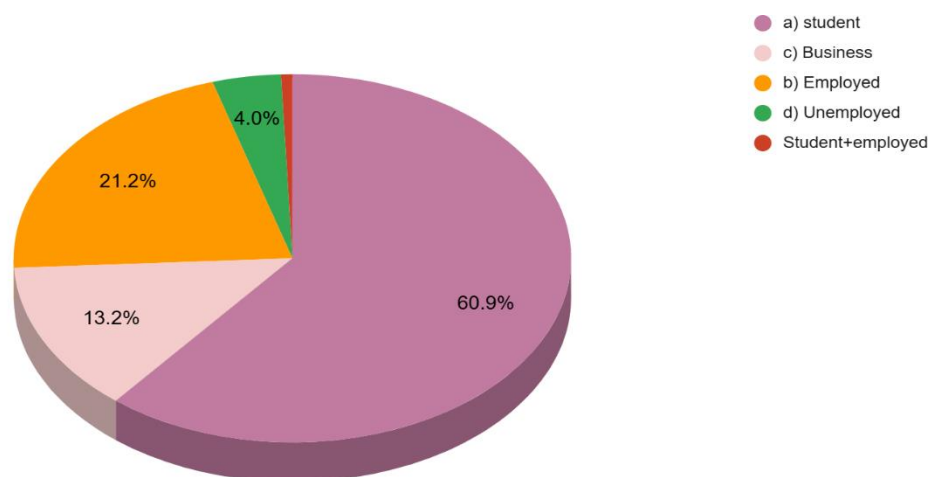
S.NO	OCCUPATION	RESPONDENTS	PERCENTAGE ANALYSIS
1	STUDENTS	92	60.9%
2	EMPLOYED	32	21.2%
3	BUSINESS	20	13.2%
4	UNEMPLOYED	6	4.0%
5	OTHER	1	0.7%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION:

The data shows that the majority of the respondents are students, making up 60.9% of the total. This is followed by employed individuals who account for 21.2%. People involved in business form 13.2% of the respondents. A smaller group, 4%, are unemployed, while only 0.7% fall under the other category.

CHART 4.4: SHOWING THE OCCUPATION OF THE RESPONDENT



**TABLE 4.5: SHOWING THE RESPONDENTS OF FREQUENTLY SHOPPING IN
ONLINE**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	DAILY	7	4.6%
2	WEEKLY	24	15.9%
3	MONTHLY	46	30.5%
4	OCCASINALLY	74	49.0%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that most respondents use the service or product only occasionally, with 49% falling into this category. About 30.5% use it on a monthly basis, while 15.9% use it weekly. Only a small number, 4.6%, use it daily.

**CHART 4.5: SHOWING THE RESPONDENTS OF FREQUENTLY SHOPPING IN
ONLINE**

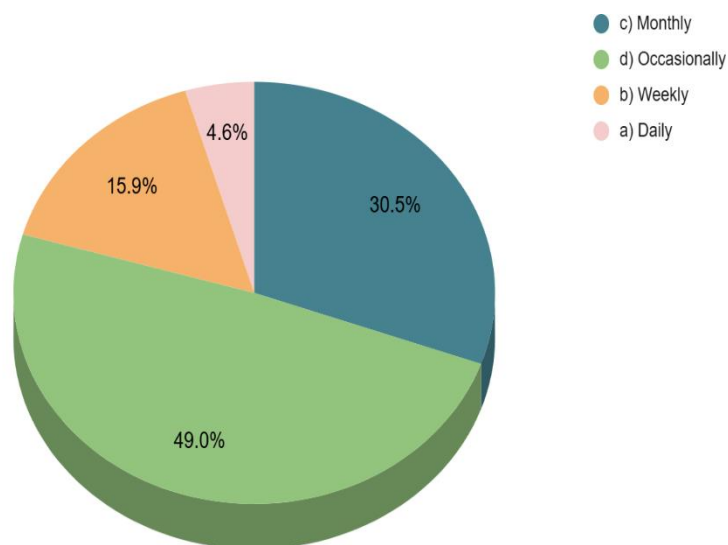


TABLE 4.6: SHOWING THE RESPONDENTS FOR THE MEMBERS IN FAMILY

S.NO	MEMBERS	RESPONDENTS	PERCENTAGE ANALYSIS
1	1	1	0.7%
2	2	5	3.3%
3	3	18	11.9%
4	4	97	64.2%
5	MORE THAN 4	30	19.9%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that most respondents, around 64.2%, come from families with 4 members. About 19.9% belong to families with more than 4 members, while 11.9% have 3 members in their family. A small percentage, 3.3%, come from families with 2 members, and only 0.7% are from single-member households.

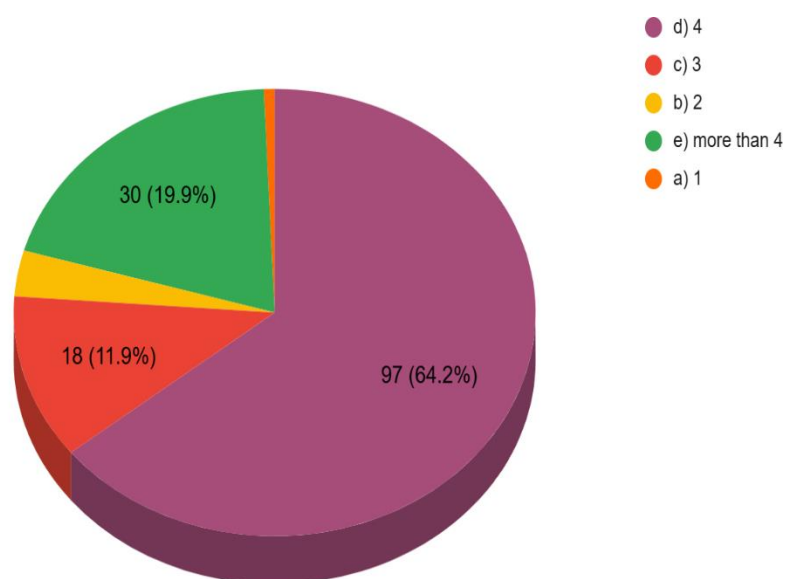
CHART 4.6: SHOWING THE RESPONDENTS FOR THE MEMBERS IN FAMILY

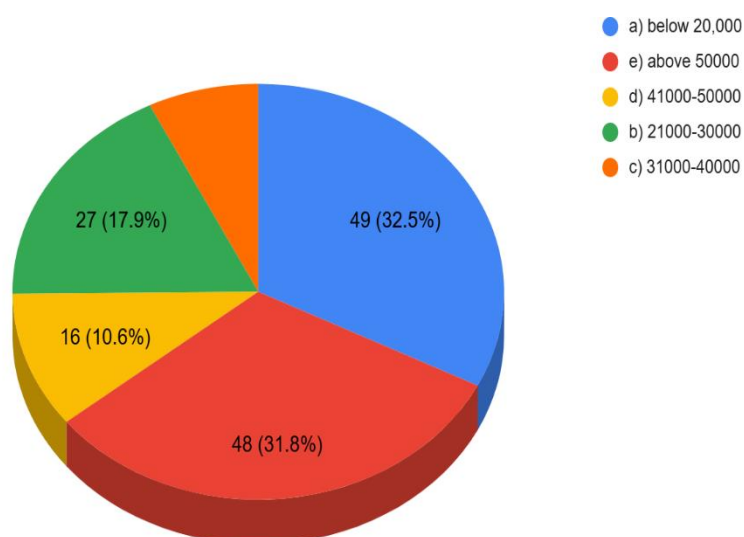
TABLE 4.7: SHOWING RESPONDENTS FOR AVERAGE MONTHLY INCOME

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	BELOW 20000	49	32.5%
2	21000-30000	27	17.9%
3	31000-40000	11	7.3%
4	41000-50000	16	10.6
5	ABOVE 50000	48	31.8%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that the largest group of respondents, 32.5%, have an average monthly income below ₹20,000. Close to this, 31.8% earn above ₹50,000 per month. Around 17.9% of the respondents have an income between ₹21,000 and ₹30,000, while 10.6% earn between ₹41,000 and ₹50,000. Only 7.3% fall in the income range of ₹31,000 to ₹40,000.

CHART 4.7: SHOWING RESPONDENTS FOR AVERAGE MONTHLY INCOME

**TABLE 4.8: SHOWING THE RESPONDENTS OF E-COMMERCE PLATFORMS
WHICH THEY USE**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	AMAZON	78	51.7
2	FLIPKART	46	30.5
3	MEESHO	10	6.6
4	OTHERS	17	11.3
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that Amazon is the most preferred e-commerce platform among respondents, with 51.7% using it. Flipkart is the second most used platform, chosen by 30.5% of the respondents. A smaller number, 6.6%, use Meesho, while 11.3% use other e-commerce platforms.

**CHART 4.8: SHOWING THE RESPONDENTS OF E-COMMERCE PLATFORMS
WHICH THEY USE**

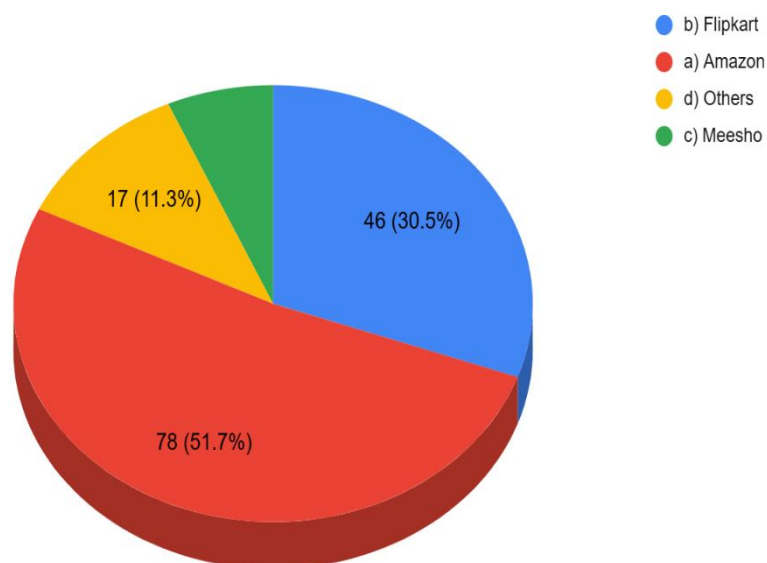


TABLE 4.9 SHOWING THE RESPONDENTS OF TYPE OF PRODUCT THEY BUY MOST IN ONLINE

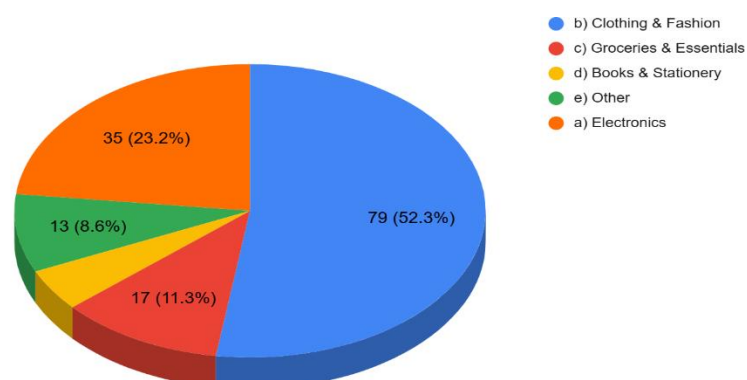
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	ELECTRONICS	35	23.2%
2	CLOTHING & FASHION	79	52.3%
3	GROCERIES & ESSENTIALS	17	11.3%
4	BOOK & STATIONERY	7	4.6
5	OTHER	13	8.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows that the majority of respondents, 52.3%, mostly buy clothing and fashion items online. Electronics are the next most purchased category, with 23.2% of respondents preferring them. About 11.3% shop for groceries and essentials, while 8.6% buy other types of products. Only 4.6% mainly purchase books and stationery online.

CHART 4.9: 9 SHOWING THE RESPONDENTS OF TYPE OF PRODUCT THEY BUY MOST IN ONLINE



**TABLE 4.10: SHOWING RESPONDENTS OF USAGE OF PAYMENT METHODS
IN ONLINE SHOPPING**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	CREDIT/DEBIT CARD	9	6.0%
2	DIGITAL WALLET(PAYPAL,GPAY, etc.)	66	43.7%
3	CASH ON DELIVERY	74	49.0%
4	BANK TRANSFER	2	1.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The above table shows how people prefer to pay while shopping online. Most of the respondents, about 49%, said they prefer Cash on Delivery. Around 43.7% of people use Digital Wallets like Google Pay, PayPal, etc., Only 6% of respondents use Credit or Debit Cards, and a very small number, just 1.3%, use Bank Transfers.

**CHART 4.10: SHOWING RESPONDENTS OF USAGE OF PAYMENT METHODS
IN ONLINE SHOPPING**

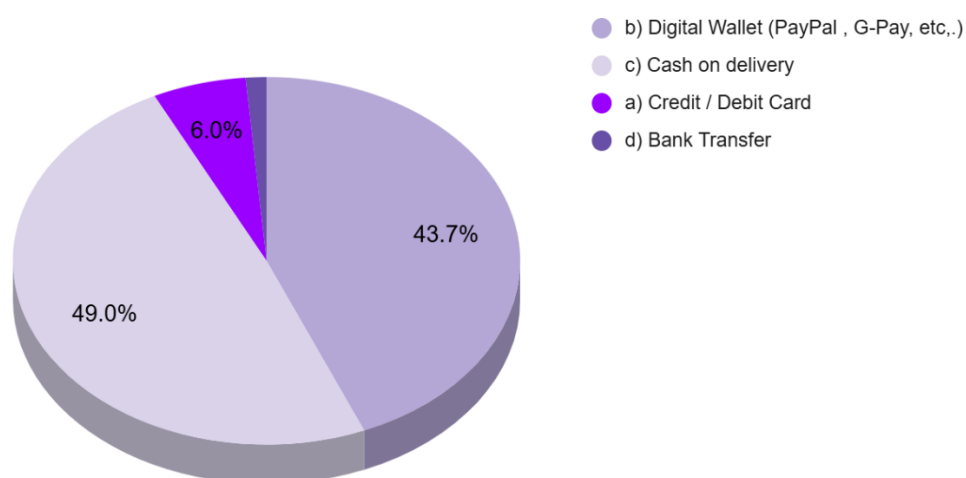


TABLE 4.11: DURATION OF ONLINE SHOPPING USAGE AMONG THE RESPONDENTS

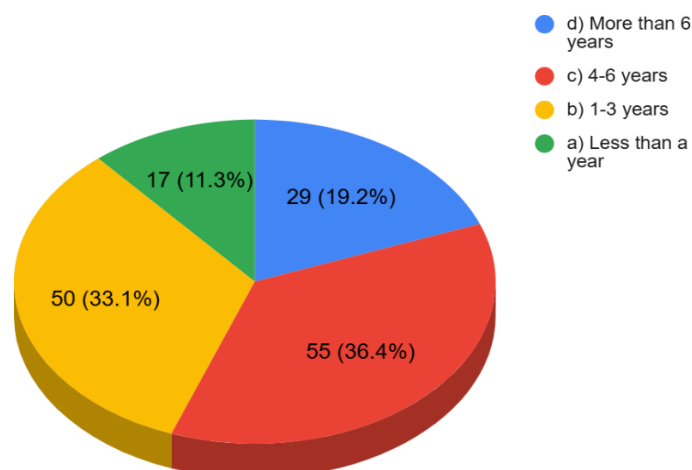
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	LESS THAN A YEAR	17	11.3%
2	1-3 YEARS	50	33.1%
3	4-6 YEARS	55	36.4%
4	MORE THAN 6 YEARS	29	19.2%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The table shows how long people have been using online shopping. Most of the respondents, around 36.4%, have been shopping online for 4 to 6 years. About 33.1% have been using it for 1 to 3 years. A smaller group, 19.2%, have been using online shopping for more than 6 years, while only 11.3% are new users who started in less than a year.

CHART 4.11: DURATION OF ONLINE SHOPPING USAGE AMONG THE RESPONDENTS



**TABLE 4.12: SHOWING THE RESPONDENTS OF USAGE OF AI-POWERED
CUSTOMER SUPPORT (e.g., chatbots) WHILE ONLINE SHOPPING**

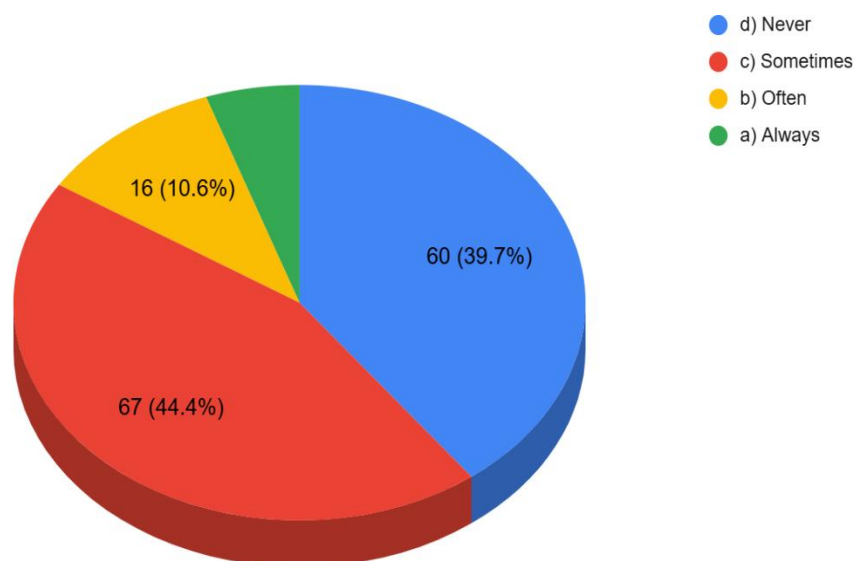
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	ALWAYS	8	5.3%
2	OFTEN	16	10.6%
3	SOMETIMES	67	44.4%
4	NEVER	60	39.7%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The table shows how often people use AI-powered customer support, like chatbots, while shopping online. The majority of respondents, about 44.4%, said they use it sometimes. Around 39.7% said they never use it. Only 10.6% said they use it often, and a very small number, just 5.3%, use it always.

**CHART 4.12: SHOWING THE RESPONDENTS OF USAGE OF AI-POWERED
CUSTOMER SUPPORT (e.g., chatbots) WHILE ONLINE SHOPPING**



**TABLE 4.13: SHOWING THE RESPONDENTS WHETHER THEY TRUST AI
RECOMMENDATIONS FOR PRODUCT SUGGESTIONS**

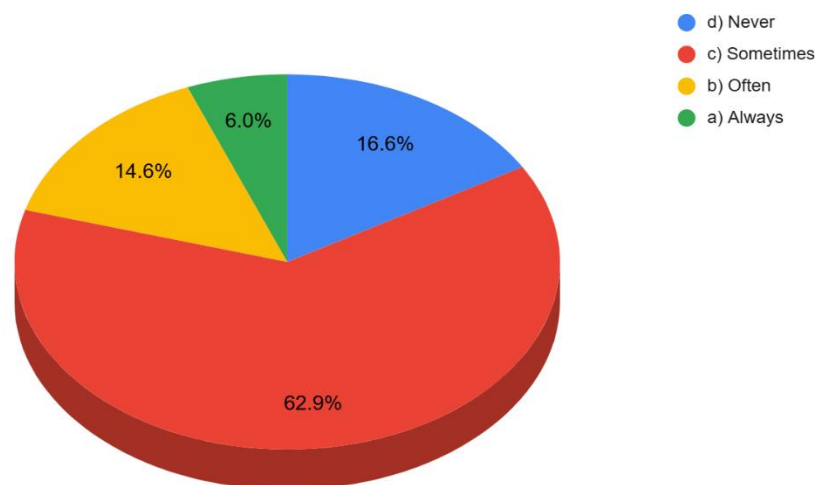
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	ALWAYS	9	6.0%
2	OFTEN	22	14.6%
3	SOMETIMES	95	62.9%
4	NEVER	25	16.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The table shows how much people trust AI recommendations for product suggestions while shopping online. Most of the respondents, about 62.9%, said they sometimes trust AI suggestions. Around 14.6% said they often trust these recommendations, and only 6% said they always trust them. On the other hand, 16.6% of respondents said they never trust AI product suggestions.

**CHART 4.13: SHOWING THE RESPONDENTS WHETHER THEY TRUST AI
RECOMMENDATIONS FOR PRODUCT SUGGESTIONS**



**TABLE 4.14: SHOWING RESPONDENTS OF USAGE OF AI-POWERED VOICE
OR IMAGE SEARCH FOR PRODUCT SEARCH**

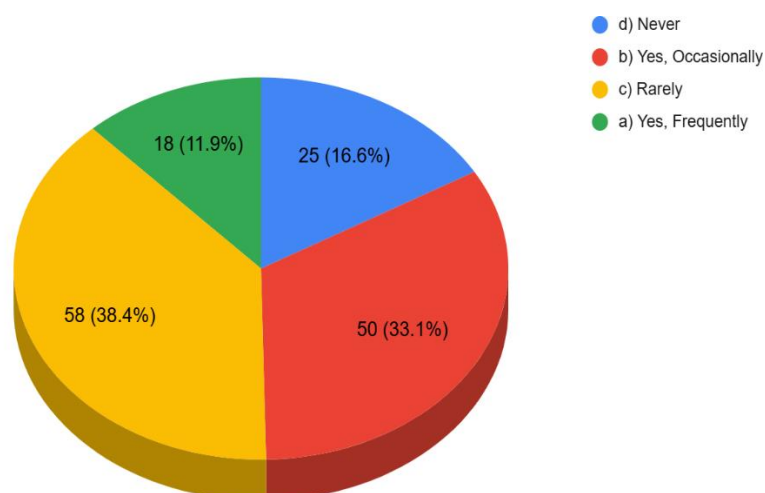
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	YES, FREQUENTLY	18	11.9%
2	YES, OCCASIONALLY	50	33.1%
3	RARELY	58	38.4%
4	NEVER	25	16.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The table shows how often people use AI-powered voice or image search to find products while shopping online. The highest number of respondents, about 38.4%, said they use it rarely. Around 33.1% use it occasionally, Only 11.9% use it frequently, which means regular use is still low. Meanwhile, 16.6% have never used voice or image search at all.

**CHART 4.14: SHOWING RESPONDENTS OF USAGE OF AI-POWERED VOICE
OR IMAGE SEARCH FOR PRODUCT SEARCH**



**TABLE 4.15: IMPORTANCE OF PERSONALIZED SHOPPING EXPERIENCE
AMONG THE RESPONDENTS**

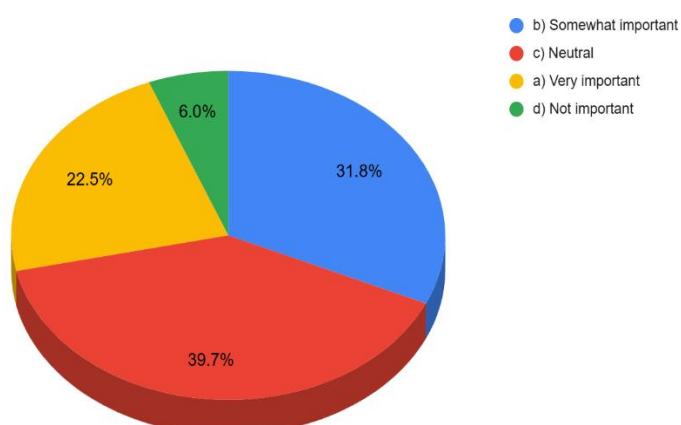
S. NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	VERY IMPORTANT	34	22.5%
2	SOMEWHAT IMPORTANT	48	31.8%
3	NEUTRAL	60	39.7%
4	NOT IMPORTANT	9	6.0%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how important a personalized shopping experience is to the respondents. Out of 151 people, 34 (22.5%) said it is very important to them, while 48 (31.8%) feel it is somewhat important. The largest group, 60 respondents (39.7%), remained neutral, meaning they neither agreed nor disagreed strongly. Only 9 people (6%) said that personalized shopping is not important to them.

**CHART 4.15: IMPORTANCE OF PERSONALIZED SHOPPING EXPERIENCE
AMONG THE RESPONDENTS**



**TABLE 4.16: SHOWING THE RESPONDENTS OF AI- POWERED
RECOMMENDATIONS HELP THEM MAKE BETTER PURCHASE DECISION**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	YES, ALWAYS	14	9.3%
2	YES, SOMETIMES	85	56.3%
3	NO, RARELY	38	25.2%
4	NO, NEVER	14	9.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI-powered recommendations helping them make better purchase decisions. Out of 151 people, the majority — 85 respondents (56.3%) — said that AI recommendations help them sometimes. Only 14 people (9.3%) felt that they always help, while 38 respondents (25.2%) said they rarely help. Another 14 people (9.3%) felt that AI recommendations never help them.

**CHART 4.16: SHOWING THE RESPONDENTS OF AI- POWERED
RECOMMENDATIONS HELP THEM MAKE BETTER PURCHASE DECISION**

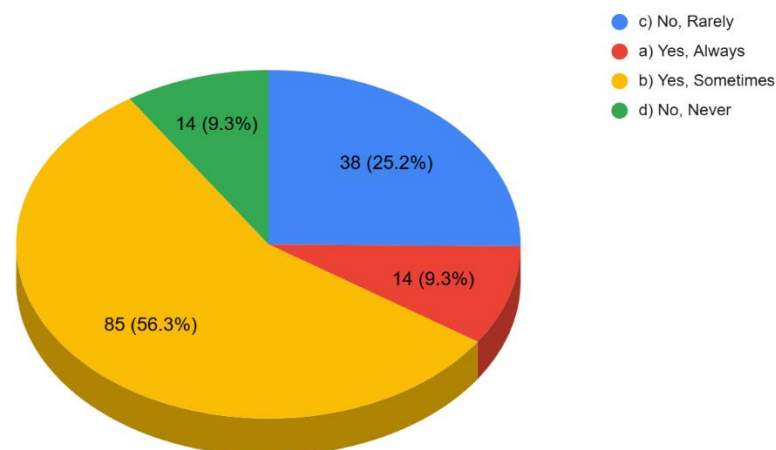


TABLE 4.17: RESPONDENTS PERCEPTION OF AI-BASED CUSTOMER SUPPORT COMPARED TO HUMAN CUSTOMER SUPPORT

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	YES, MUCH BETTER	22	14.6%
2	YES, SOMEWHAT BETTER	56	37.1%
3	NO DIFFERENCE	31	20.5%
4	NO, HUMAN SUPPORT IS BETTER	42	27.8%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how people feel about AI-based customer support compared to human customer support. Out of 151 respondents, 22 people (14.6%) said AI support is much better, and 56 (37.1%) felt it is somewhat better. Meanwhile, 31 respondents (20.5%) felt there is no real difference between the two. However, 42 people (27.8%) believed that human support is still better than AI.

CHART: 4.17: RESPONDENTS PERCEPTION OF AI-BASED CUSTOMER SUPPORT COMPARED TO HUMAN CUSTOMER SUPPORT

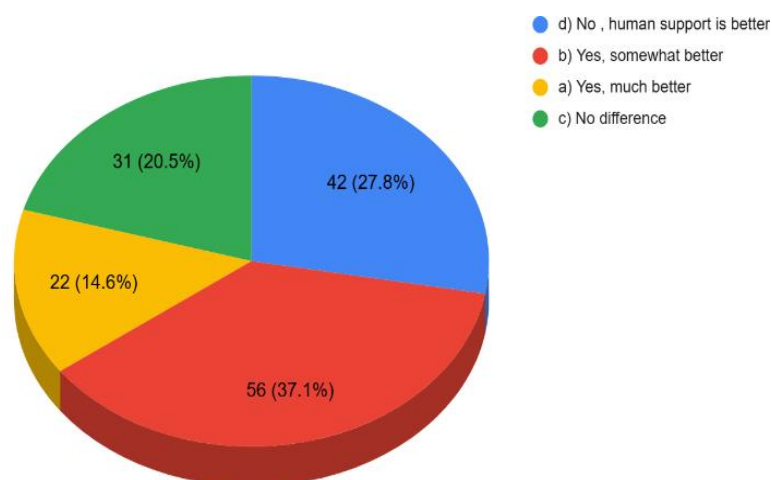


TABLE 4.18: RESPONDENTS LEVEL OF CONCERN REGARDING AI-BASED SECURITY AND PRIVACY IN ONLINE SHOPPING

S.NO	USGAE	RESPONDENTS	PERCENTAGE ANALYSIS
1	VERY CONCERNED	22	14.6%
2	SOMEWHAT CONCERNED	98	64.9%
3	NEUTRAL	25	16.6%
4	NOT CONCERNED	6	4.0%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how concerned people are about AI-based security and privacy in online shopping. Out of 151 respondents, the majority — 98 people (64.9%) — said they are somewhat concerned, while 22 people (14.6%) are very concerned. About 25 respondents (16.6%) were neutral, showing neither concern nor comfort. Only 6 people (4%) said they are not concerned at all.

CHART 4.18: SHOWING THE RESPONDENTS OF HOW THEY CONCERNED ABOUT AI-BASED SECURITY AND PRIVACY IN ONLINE SHOPPING

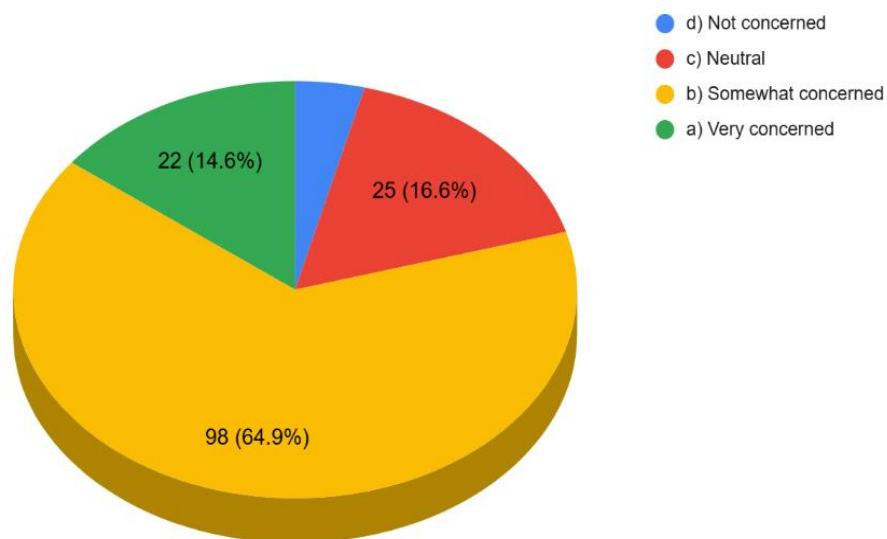


TABLE 4.19: RESPONDENTS TRUST IN ONLINE PAYMENTS DUE TO AI-DRIVEN FRAUD DETECTION SYSTEMS

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	YES, SIGNIFICANTLY	32	21.2%
2	YES, SOMEWHAT	71	47.0%
3	NO, IT DOES NOT MATTER	31	20.5%
4	NO, I DON'T TRUST IT	17	11.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how AI-driven fraud detection systems affect people's trust in online payments. Out of 151 respondents, 71 people (47.0%) said these systems somewhat increase their trust, and 32 people (21.2%) said they significantly increase their trust. On the other hand, 31 respondents (20.5%) said it doesn't make a difference to them, while 17 people (11.3%) said they don't trust it at all.

CHART 4.19: RESPONDENTS TRUST IN ONLINE PAYMENTS DUE TO AI-DRIVEN FRAUD DETECTION SYSTEMS

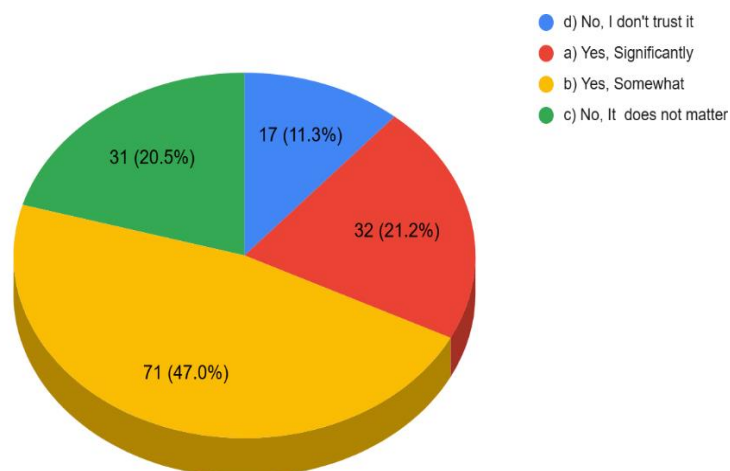


TABLE 4.20: RESPONDENTS EXPERIENCE WITH AI-DRIVEN SECURITY FEATURES (e.g., AUTOMATED FRAUD DETECTION) IN ONLINE SHOPPING

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	YES,FREQUENTLY	19	12.6%
2	YES, OCCASIONALLY	40	26.5%
3	RARELY	48	31.8%
4	NEVER	44	29.1%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how often respondents have experienced AI-driven security features, such as automated fraud detection, during online shopping. Out of 151 people, 40 (26.5%) said they have experienced it occasionally, while 19 (12.6%) said they experience it frequently. A larger group of 48 respondents (31.8%) said they have rarely come across such features, and 44 people (29.1%) said they have never experienced them.

CHART 4.20: RESPONDENTS EXPERIENCE WITH AI-DRIVEN SECURITY FEATURES (e.g., AUTOMATED FRAUD DETECTION) IN ONLINE SHOPPING

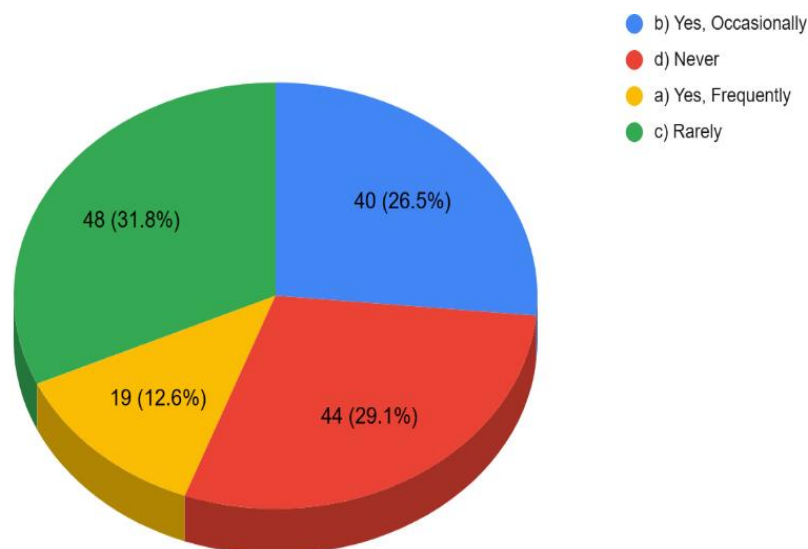


TABLE 4.21: RESPONDENTS LIKELIHOOD OF RECOMMENDING AI-DRIVEN E-COMMERCE PLATFORMS BASED ON THEIR EXPERIENCE

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	YES, DEFINITELY	24	15.9%
2	YES, PROBABLY	59	39.1%
3	NOT SURE	57	37.7%
4	NO	11	7.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how likely respondents are to recommend AI-driven e-commerce platforms to others based on their experience. Out of 151 people, 59 (39.1%) said they would probably recommend them, and 24 (15.9%) said they would definitely recommend them. However, 57 respondents (37.7%) were not sure, and 11 people (7.3%) said they would not recommend them.

CHART 4.21: RESPONDENTS LIKELIHOOD OF RECOMMENDING AI-DRIVEN E-COMMERCE PLATFORMS BASED ON THEIR EXPERIENCE

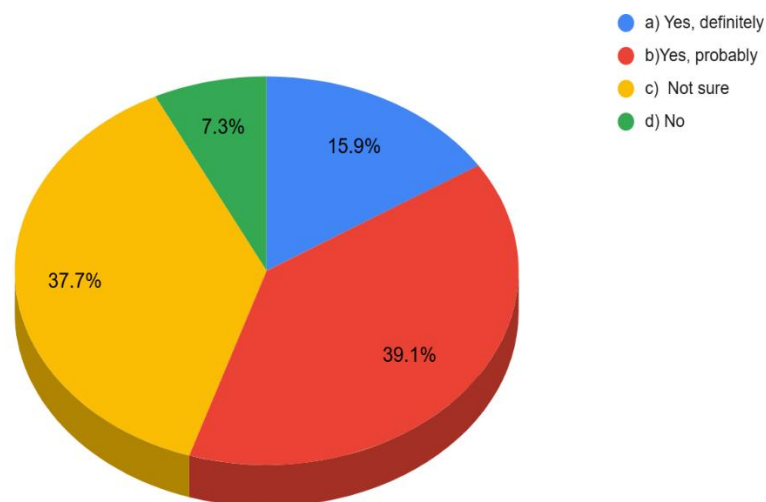


TABLE 4.22: HOW AI IN CUSTOMER SUPPORT AND SERVICES QUALITY

TABLE 4.22.1: RESPONDENTS VIEWS ON THE EFFECTIVENESS AND SPEED OF AI-POWERED CHATBOTS IN HANDLING CUSTOMER QUERIES

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	18	11.9%
2	DISAGREE	21	13.9
3	NETURAL	48	31.8%
4	AGREE	50	33.1%
5	STRONGLY AGREE	14	9.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how people feel about AI-powered chatbots giving quick and effective answers to customer questions. Around one-third of the respondents (33.1%) agree that chatbots are helpful in this way, and 9.3% strongly agree. However, 31.8% of people are neither agree nor disagree. On the other hand, 13.9% disagree and 11.9% strongly disagree, meaning a quarter of the respondents are not satisfied with the performance of chatbots.

CHART 4.22.1: RESPONDENTS VIEWS ON THE EFFECTIVENESS AND SPEED OF AI-POWERED CHATBOTS IN HANDLING CUSTOMER QUERIES

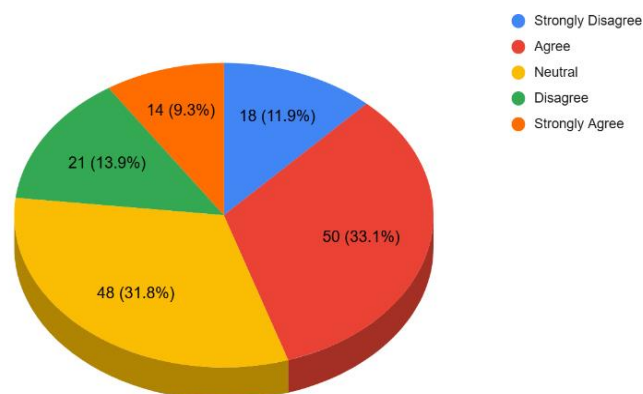


TABLE 4.22.2: RESPONDENTS PERCEPTION OF HOW AI-BASED VIRTUAL ASSISTANTS IMPROVE THE EFFICIENCY OF CUSTOMER SERVICE

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	12	7.9%
2	DISAGREE	22	14.6%
3	NEUTRAL	50	33.1%
4	AGREE	58	38.4%
5	STRONGLY AGREE	9	6.0%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how people feel about AI-based virtual assistants improving customer service efficiency. A significant number of respondents (38.4%) agree that these assistants help make customer service more efficient, and 6% strongly agree. However, a large portion (33.1%) remain neither agree nor disagree, On the other side, 14.6% disagree and 7.9% strongly disagree, suggesting that some users are not satisfied with the efficiency of AI-based assistants.

CHART4.22.2: RESPONDENTS VIEWS ON THE EFFECTIVENESS AND SPEED OF AI-POWERED CHATBOTS IN HANDLING CUSTOMER QUERIES

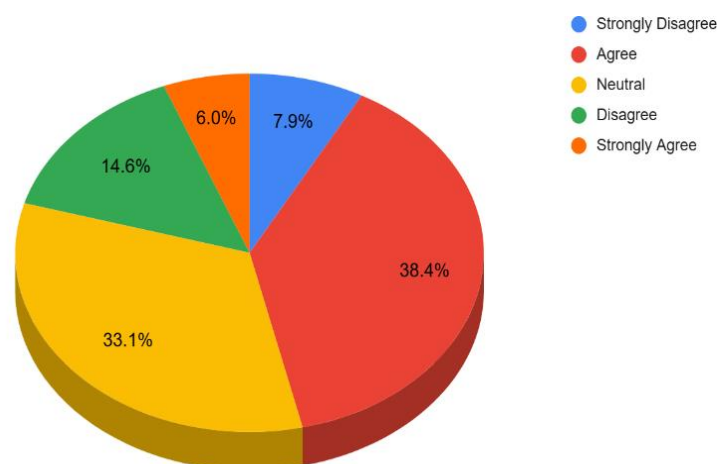


TABLE 4.22.3: RESPONDENTS VIEWS ON HOW AI REDUCES CUSTOMER WAITING TIME THROUGH AUTOMATED RESPONSES AND SOLUTIONS

S.NO	USGAE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	11	7.3%
2	DISAGREE	16	10.6%
3	NEUTRAL	54	35.8%
4	AGREE	47	31.1%
5	STRONGLY AGREE	23	15.2%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data reflects people's opinions on how AI helps reduce customer waiting time by automating responses and solutions. About 31.1% of the respondents agree that AI helps speed up the process, and 15.2% strongly agree, However, 35.8% are neither agree nor disagree, meaning they are either unsure or haven't noticed a major difference. On the other hand, a smaller portion—10.6% disagree and 7.3% strongly disagree—feel that AI hasn't helped reduce waiting times.

CHART 4.22.3: RESPONDENTS VIEWS ON HOW AI REDUCES CUSTOMER WAITING TIME THROUGH AUTOMATED RESPONSES AND SOLUTIONS

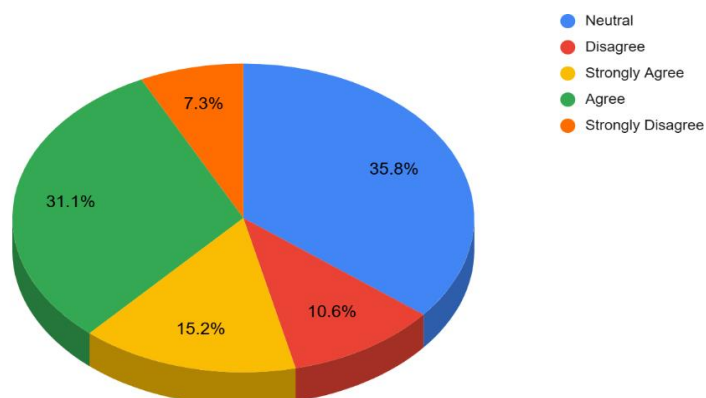


TABLE 4.22.4: RESPONDENTS PERCEPTION OF HOW AI-DRIVEN CUSTOMER SUPPORT ENHANCES PROBLEM RESOLUTION ACCURACY

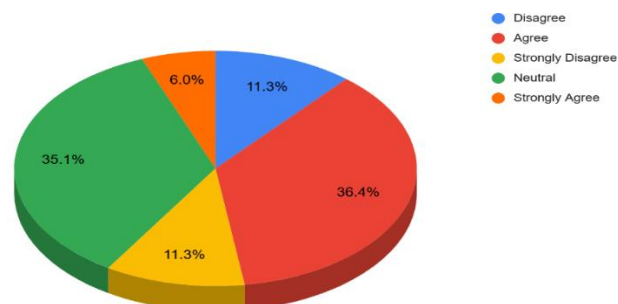
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	17	11.3%
2	DISAGREE	17	11.3%
3	NEUTRAL	53	35.1%
4	AGREE	55	36.4%
5	STRONGLY AGREE	9	6.0%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows what people think about how well AI-driven customer support improves the accuracy of solving problems. A good number of respondents (36.4%) agree that AI helps resolve issues more accurately, and 6% strongly agree. However, a large portion—35.1%—are neither agree nor disagree, On the other hand, 11.3% strongly disagree and another 11.3% disagree, showing that some people are not confident in AI's ability to accurately solve problems.

CHART 4.22.4: RESPONDENTS PERCEPTION OF HOW AI-DRIVEN CUSTOMER SUPPORT ENHANCES PROBLEM RESOLUTION ACCURACY



**TABLE 4.22.5: RESPONDENTS PERCEPTION OF HOW AI IMPROVES THEIR
OVERALL EXPERIENCE WITH E-COMMERCE CUSTOMER SUPPORT
SERVICES.**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	11	7.3%
2	DISAGREE	25	16.6%
3	NEUTRAL	64	42.4%
4	AGREE	38	25.2%
5	STRONGLY AGREE	13	8.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how people feel about the overall impact of AI on their experience with e-commerce customer support services. About 25.2% of the respondents agree that AI has improved their experience, and 8.6% strongly agree. However, the majority—42.4%—remain neither agree nor disagree, On the negative side, 16.6% disagree and 7.3% strongly disagree, meaning some users feel that AI has not improved their experience.

**CHART 4.22.5: RESPONDENTS PERCEPTION OF HOW AI IMPROVES THEIR
OVERALL EXPERIENCE WITH E-COMMERCE CUSTOMER SUPPORT
SERVICES.**

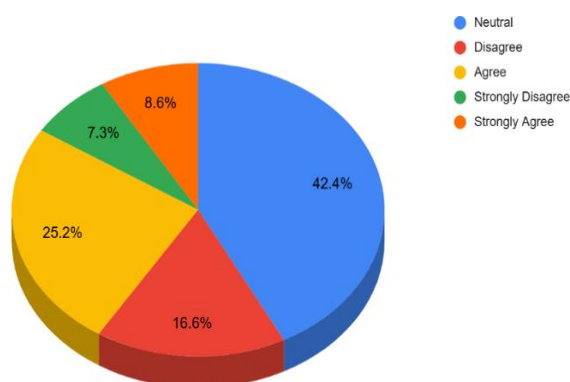


TABLE 4.23: AI-DRIVEN PERSONALIZATION AND CUSTOMER EXPERIENCE

TABLE 4.23.1: RESPONDENTS PERCEPTION OF THE RELEVANCE AND USEFULNESS OF AI-POWERED PRODUCT RECOMMENDATIONS

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	26	17.2%
2	DISAGREE	20	13.2%
3	NEUTRAL	52	34.4%
4	AGREE	46	30.5%
5	STRONGLY AGREE	7	4.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data in Table 4.23.1 shows how respondents feel about the relevance and usefulness of AI-powered product recommendations. A majority of the participants were either neither agree nor disagree (34.4%) or agreed (30.5%) that these recommendations are helpful. However, a smaller portion strongly agreed (4.6%). On the other hand, with 17.2% strongly disagreeing and 13.2% disagreeing.

CHART 4.23.1: RESPONDENTS PERCEPTION OF THE RELEVANCE AND USEFULNESS OF AI-POWERED PRODUCT RECOMMENDATIONS

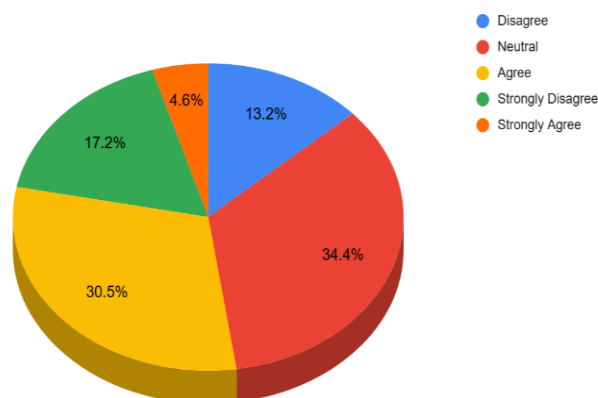


TABLE 4.23.2: RESPONDENTS PERCEPTION OF HOW AI-BASED TARGETED PROMOTIONS AND DISCOUNTS ENCOURAGE INCREASED PURCHASING

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	11	7.3%
2	DISAGREE	34	22.5%
3	NEUTRAL	53	35.1%
4	AGREE	43	28.5%
5	STRONGLY AGREE	10	6.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The table shows how respondents feel about AI-based targeted promotions and discounts and whether they encourage more purchases. Most respondents were neither agree nor disagree (35.1%). However, 28.5% agreed and 6.6% strongly agreed that these AI-driven offers encouraged them to buy more. On the other hand, 22.5% disagreed and 7.3% is strongly disagree.

CHART 4.23.2: RESPONDENTS PERCEPTION OF HOW AI-BASED TARGETED PROMOTIONS AND DISCOUNTS ENCOURAGE INCREASED PURCHASING

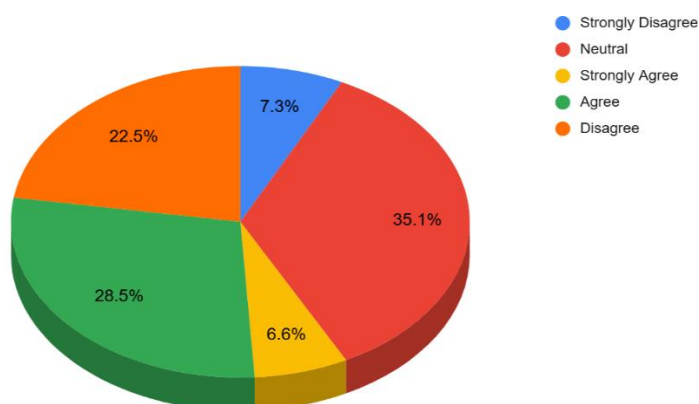


TABLE 4.23.3: RESPONDENTS PERCEPTION OF HOW AI CUSTOMIZES THEIR SHOPPING JOURNEY BASED ON BROWSING AND PURCHASE HISTORY

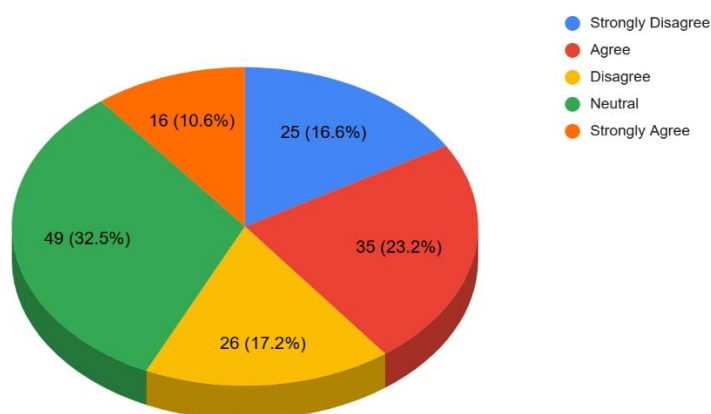
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	25	16.6%
2	DISAGREE	26	17.2%
3	NEUTRAL	49	32.5%
4	AGREE	35	23.2%
5	STRONGLY AGREE	16	10.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data in Table 4.23.3 highlights how respondents feel about AI customizing their shopping journey based on their browsing and purchase history. A significant portion of respondents were neither agree nor disagree (32.5%), Meanwhile, 23.2% agreed and 10.6% strongly agreed that AI does personalize their shopping experience. On the other hand, 17.2% disagreed and 16.6% strongly disagreed, suggesting that some users did not notice or believe in such customization.

CHART 4.23.3: RESPONDENTS PERCEPTION OF HOW AI CUSTOMIZES THEIR SHOPPING JOURNEY BASED ON BROWSING AND PURCHASE HISTORY



**TABLE 4.23.4: RESPONDENTS PERCEPTION OF HOW AI HELPS THEM
DISCOVER NEW PRODUCTS THAT MATCH THEIR PREFERENCES**

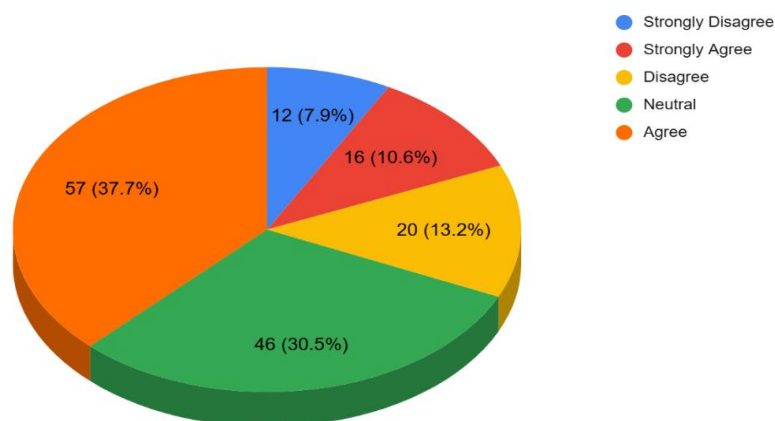
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	12	7.9%
2	DISAGREE	20	13.2%
3	NEUTRAL	46	30.5%
4	AGREE	57	37.7%
5	STRONGLY AGREE	16	10.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI helping them discover new products that match their preferences. The majority of respondents had a positive view, with 37.7% agreeing and 10.6% strongly agreeing. Around 30.5% were neither agree nor disagree, On the other hand, 13.2% disagreed and 7.9% strongly disagreed, suggesting that a smaller group did not find AI helpful in this area.

**CHART 4.23.4: RESPONDENTS PERCEPTION OF HOW AI HELPS THEM
DISCOVER NEW PRODUCTS THAT MATCH THEIR PREFERENCES**



**TABLE4.23.5: RESPONDENTS PERCEPTION OF HOW AI-DRIVEN
PERSONALIZATION IMPROVES THEIR SHOPPING EXPERIENCE**

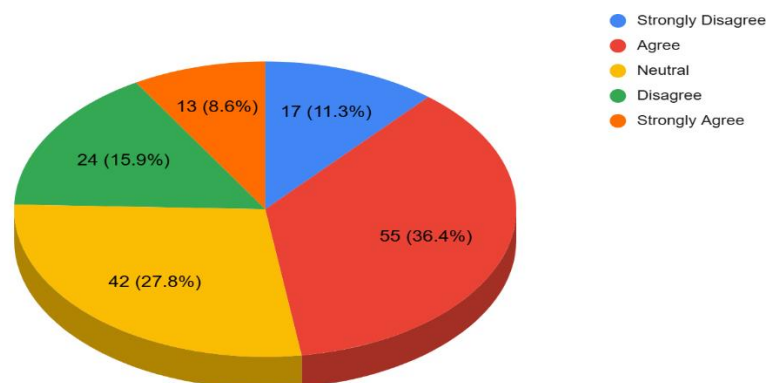
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	17	11.3%
2	DISAGREE	24	15.9%
3	NEUTRAL	42	27.8%
4	AGREE	55	36.4%
5	STRONGLY AGREE	13	8.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI-driven personalization improving their shopping experience. A large number of respondents had a positive opinion, with 36.4% agreeing and 8.6% strongly agreeing that it enhances their experience. About 27.8% were neither agree nor disagree, Meanwhile, 15.9% disagreed and 11.3% strongly disagreed, indicating that some users did not feel any improvement.

**CHART4.23.5: RESPONDENTS PERCEPTION OF HOW AI-DRIVEN
PERSONALIZATION IMPROVES THEIR SHOPPING EXPERIENCE**



**TABLE 4.24: AI IN CUSTOMER ENGAGEMENT ON E-COMMERCE
PLATFORMS**

**TABLE 4.24.1: RESPONDENTS PERCEPTION OF HOW AI-POWERED
CHATBOTS MAKE THEIR SHOPPING EXPERIENCE MORE INTERACTIVE**

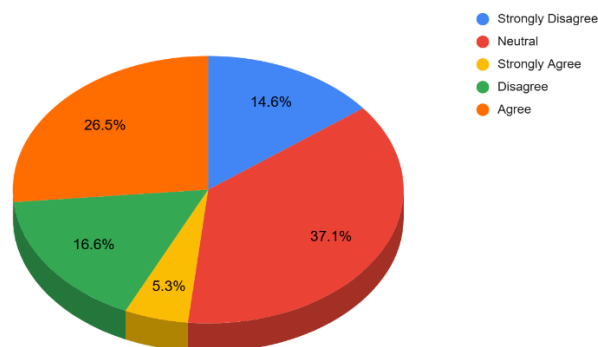
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	22	14.6%
2	DISAGREE	25	16.6%
3	NEUTRAL	56	37.1%
4	AGREE	40	26.5%
5	STRONGLY AGREE	8	5.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how people feel about the use of AI-powered chatbots in making their shopping experience more interactive. A small number of respondents strongly agreed (5.3%) and 26.5% agreed that chatbots improve their experience. However, a larger group (37.1%) remained neutral, neither agree nor disagree. On the other hand, 14.6% strongly disagreed and 16.6% disagreed, showing that about one-third of the respondents did not find chatbots useful.

**CHART4.24.1: RESPONDENTS PERCEPTION OF HOW AI-POWERED
CHATBOTS MAKE THEIR SHOPPING EXPERIENCE MORE INTERACTIVE**



**TABLE4.24.2: RESPONDENTS PERCEPTION OF HOW AI DRIVEN
PERSONALIZED NOTIFICATIONS KEEP THEM ENGAGED WITH THE
PLATFORM**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	17	11.3
2	DISAGREE	25	16.6%
3	NEUTRAL	50	33.1%
4	AGREE	47	31.1%
5	STRONGLY AGREE	12	7.9%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The table shows how users feel about AI-driven personalized notifications and whether these keep them engaged with the platform. About 31.1% of respondents agreed and 7.9% strongly agreed, meaning nearly 39% believe that personalized notifications help keep them engaged. On the other hand, 33.1% of the respondents remained neither agree nor disagree, Meanwhile, 16.6% disagreed and 11.3% strongly disagreed, showing that around 28% do not find such notifications engaging.

**CHART4.24.2: RESPONDENTS PERCEPTION OF HOW AI DRIVEN
PERSONALIZED NOTIFICATIONS KEEP THEM ENGAGED WITH THE
PLATFORM**

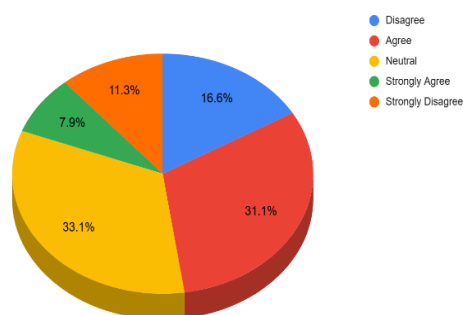


TABLE 4.24.3: SHOWING THE RESPONDENTS OF AI-POWERED VOICE AND VISUAL SEARCH MAKE IT EASIER TO FIND PRODUCTS.

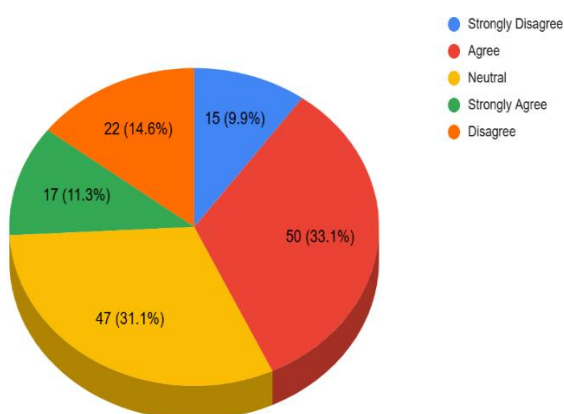
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	15	9.9%
2	DISAGREE	22	14.6%
3	NEUTRAL	47	31.1%
4	AGREE	50	33.1%
5	STRONGLY AGREE	17	11.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about the usefulness of AI-powered voice and visual search in making it easier to find products. About 33.1% agreed and 11.3% strongly agreed, meaning around 44.4% of the respondents found these features helpful in improving their product search experience. On the other hand, 31.1% remained neither agree nor disagree, A smaller group, 9.9% strongly disagreed and 14.6% disagreed (totaling 24.5%), did not find voice and visual search useful.

CHART 4.24.3: SHOWING THE RESPONDENTS OF AI-POWERED VOICE AND VISUAL SEARCH MAKE IT EASIER TO FIND PRODUCTS.



**TABLE 4.24.4: SHOWING THE RESPONDENTS OF AI ENHANCES THEM
ENGAGEMENT BY OFFERING CUSTOMIZED DEALS AND RECOMMENDATIONS.**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	11	7.3%
2	DISAGREE	23	15.2%
3	NEUTRAL	47	31.1%
4	AGREE	61	40.4%
5	STRONGLY AGREE	9	6.0%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI enhancing their engagement by offering customized deals and recommendations. A significant portion, 40.4% agreed and 6.0% strongly agreed, meaning nearly 46.4% of the respondents believe that personalized offers and suggestions from AI help keep them engaged on the platform. Meanwhile, 31.1% stayed neither agree nor disagree, On the other hand, 15.2% disagreed and 7.3% strongly disagreed, suggesting that around 22.5% do not feel that AI-driven customization improves their engagement.

**CHART 4.24.4: SHOWING THE RESPONDENTS OF AI ENHANCES THEM
ENGAGEMENT BY OFFERING CUSTOMIZED DEALS AND RECOMMENDATIONS.**

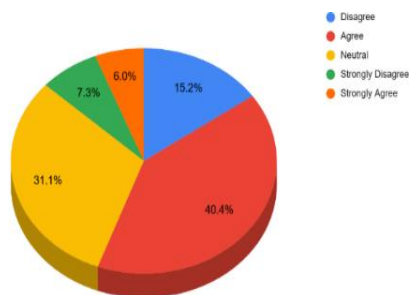


TABLE 4.24.5: SHOWING THE RESPONDENTS OF AI-DRIVEN MARKETING STRATEGIES (EMAILS, MESSAGES, ETC.) KEEP THEIR CONNECTED TO E-COMMERCE PLATFORMS.

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	23	15.2%
2	DISAGREE	19	12.6%
3	NEUTRAL	50	33.1
4	AGREE	43	28.5%
5	STRONGLY AGREE	16	10.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI-driven marketing strategies, such as emails and messages, in keeping them connected to e-commerce platforms. Around 28.5% agreed and 10.6% strongly agreed, which means about 39.1% of respondents feel that these AI-based strategies help maintain their connection with online shopping platforms. Meanwhile, 33.1% remained neither agree nor disagree, On the other hand, 15.2% strongly disagreed and 12.6% disagreed, making up 27.8% who do not feel that these strategies are effective.

CHART 4.24.5: SHOWING THE RESPONDENTS OF AI-DRIVEN MARKETING STRATEGIES (EMAILS, MESSAGES, ETC.) KEEP THEIR CONNECTED TO E-COMMERCE PLATFORMS.

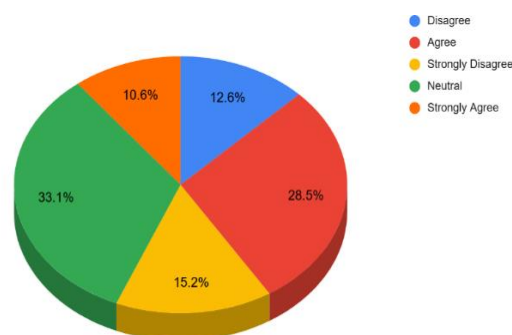


TABLE 4.25: AI-BASED SAFETY AND SECURITY FEATURES ON CUSTOMER TRUST AND SATISFACTION

TABLE 4.25.1: RESPONDENTS PERCEPTION OF HOW AI HELPS IN DETECTING FRAUDULENT ACTIVITIES ON E-COMMERCE PLATFORMS

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	24	15.9%
2	DISAGREE	17	11.3%
3	NEUTRAL	60	39.7%
4	AGREE	40	26.5%
5	STRONGLY AGREE	10	6.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents view the role of AI in detecting fraudulent activities on e-commerce platforms. About 26.5% agreed and 6.6% strongly agreed, meaning around 33.1% believe that AI helps in identifying fraud. However, the largest portion of respondents, 39.7%, remained neither agree nor disagree, suggesting that many are unsure about how effective AI is in this area. On the other hand, 15.9% strongly disagreed and 11.3% disagreed, totaling 27.2% who do not think AI is effective in detecting fraud.

CHART 4.25.1: RESPONDENTS PERCEPTION OF HOW AI HELPS IN DETECTING FRAUDULENT ACTIVITIES ON E-COMMERCE PLATFORMS

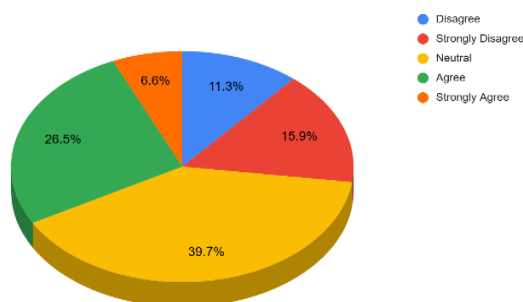


TABLE4.25.2: SHOWING THE RESPONDENTS OF AI-BASED SECURITY MEASURES INCREASE THEIR TRUST IN ONLINE SHOPPING.

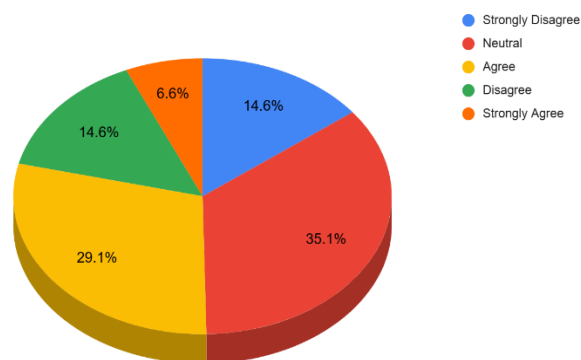
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	22	14.6%
2	DISAGREE	22	14.6%
3	NEUTRAL	53	35.1%
4	AGREE	44	29.1%
5	STRONGLY AGREE	10	6.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI-based security measures and whether they increase trust in online shopping. Around 29.1% agreed and 6.6% strongly agreed, meaning about 35.7% of respondents feel that these security features help build their trust in e-commerce platforms. Meanwhile, 35.1% remained neither agree nor disagr, On the other hand, 14.6% strongly disagreed and another 14.6% disagreed, making up 29.2% who do not feel that AI-based security increases their trust.

CHART4.25.2: SHOWING THE RESPONDENTS OF AI-BASED SECURITY MEASURES INCREASE THEIR TRUST IN ONLINE SHOPPING.



**TABLE 4.25.3: SHOWING THE RESPONDENTS OF AI PREVENTS
UNAUTHORIZED ACCESS TO THEIR E-COMMERCE ACCOUNTS.**

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	14	9.3%
2	DISAGREE	9	6.0%
3	NEUTRAL	60	39.7%
4	AGREE	44	29.1%
5	STRONGLY AGREE	24	15.9%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about the role of AI in preventing unauthorized access to their e-commerce accounts. About 29.1% agreed and 15.9% strongly agreed, which means a total of 45% believe that AI helps protect their accounts from unauthorized access. A significant portion, 39.7%, remained neither agree nor disagree, On the other hand, 9.3% strongly disagreed and 6.0% disagreed, totaling 15.3% who do not think AI effectively prevents unauthorized access.

**CHART4.25.3: SHOWING THE RESPONDENTS OF AI PREVENTS
UNAUTHORIZED ACCESS TO THEIR E-COMMERCE ACCOUNTS.**

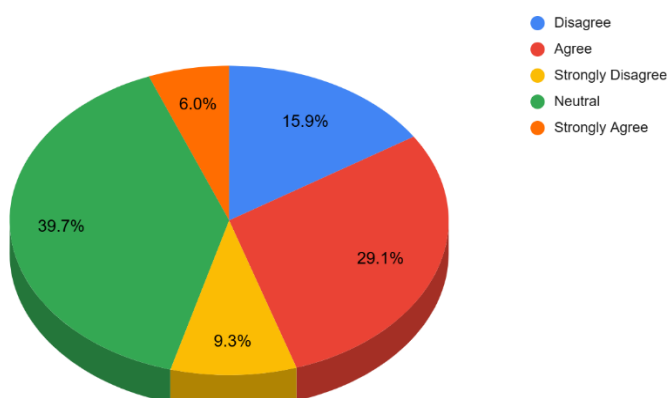


TABLE 4.25.4: SHOWING THE RESPONDENTS OF AI-POWERED IDENTITY VERIFICATION ENHANCES THE SECURITY OF ONLINE TRANSACTIONS.

S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	12	7.9%
2	DISAGREE	22	14.6
3	NEUTRAL	59	39.1%
4	AGREE	50	33.1%
5	STRONGLY AGREE	8	5.3%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI-powered identity verification and its role in enhancing the security of online transactions. Around 33.1% agreed and 5.3% strongly agreed, meaning a total of 38.4% believe that this technology improves the safety of their online transactions. However, the largest group—39.1%—remained neither agree nor disagree, On the other hand, 14.6% disagreed and 7.9% strongly disagreed, totaling 22.5% who do not think AI-powered identity verification significantly boosts transaction security.

CHART4.25.4: SHOWING THE RESPONDENTS OF AI-POWERED IDENTITY VERIFICATION ENHANCES THE SECURITY OF ONLINE TRANSACTIONS.

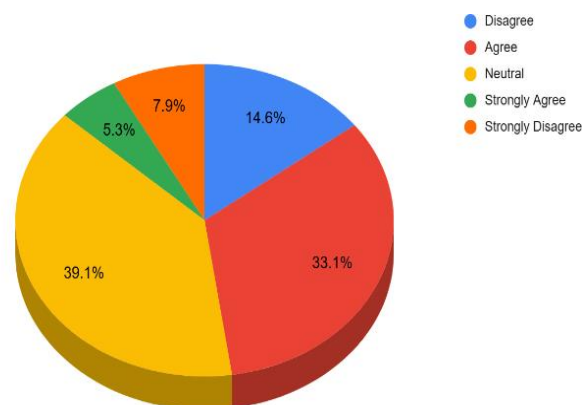


TABLE 4.25.5: SHOWING THE RESPONDENTS OF AI-DRIVEN SECURITY FEATURES GIVE THEM CONFIDENCE IN MAKING ONLINE PURCHASES.

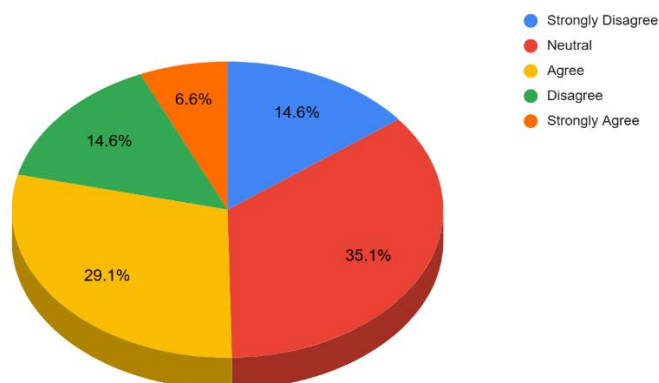
S.NO	USAGE	RESPONDENTS	PERCENTAGE ANALYSIS
1	STRONGLY DISAGREE	22	14.6%
2	DISAGREE	22	14.6%
3	NEUTRAL	53	35.1%
4	AGREE	44	29.1%
5	STRONGLY AGREE	10	6.6%
	TOTAL	151	100

Source: Primary Data

INTERPRETATION

The data shows how respondents feel about AI-driven security features giving them confidence in making online purchases. Around 29.1% agreed and 6.6% strongly agreed, meaning a total of 35.7% of respondents feel that these features make them more confident while shopping online. Meanwhile, 35.1% remained neither agree nor disagree, On the other hand, 14.6% strongly disagreed and another 14.6% disagreed, making up 29.2% who do not feel that AI-based security features increase their trust.

CHART 4.25.5: SHOWING THE RESPONDENTS OF AI-DRIVEN SECURITY FEATURES GIVE THEM CONFIDENCE IN MAKING ONLINE PURCHASES.



4.26. CHI-SQUARE TEST: CROSS TABULATION OF GENDER AND USAGE OF AI-POWERED CUSTOMER SUPPORT (e.g., chatbots) WHILE ONLINE SHOPPING

NULL HYPOTHESIS (H₀):

There is no significant association between gender of respondents and their trust in online payments due to AI-driven fraud detection systems.

ALTERNATIVE HYPOTHESIS (H₁):

There is a significant association between gender of respondents and their trust in online payments due to AI-driven fraud detection systems.

1.OBSERVED OUTCOME

	NEVER	SOMETIMES	ALWAYS	OFTEN	
MALE	16	31	4	9	60
FEMALE	44	36	4	7	91
	60	67	8	16	151

EXPECTED OUTCOME

	NEVER	SOMETIMES	ALWAYS	OFTEN
MALE	23.8410596	26.62251656	3.178808	6.357616
FEMALE	36.1589404	40.37748344	4.821192	9.642384

(O-E)²/E

	NEVER	SOMETIMES	ALWAYS	OFTEN
MALE	2.57883738	0.719780238	0.212141	1.098241
FEMALE	1.700332339	0.474580377	0.139873	0.724115

X ²	7.647900755
DF	1
P-VALUE	0.005683842
CHI-SQUARE	0.053877583

INTERPRETATION:

The Chi-Square test was used to check whether there is a relationship between gender (male and female) and the type of responses (Never, Sometimes, Always, Often). The calculated Chi-Square value is 7.648 and the p-value is 0.054. Since the p-value is greater than 0.05, the result is not statistically significant. This means the null hypothesis is accepted.

4.26.1: CHI-SQUARE TEST: CROSS TABULATION OF GENDER AND TRUST IN ONLINE PAYMENTS DUE TO AI-DRIVEN FRAUD DETECTION SYSTEMS

H₀ (NULL HYPOTHESIS):

There is no significant association between gender and trust in online payments due to AI-driven fraud detection systems. (Gender does not affect the trust level.)

H₁ (ALTERNATIVE HYPOTHESIS):

There is a significant association between gender and trust in online payments due to AI-driven fraud detection systems.

OBSERVED OUTCOME

	YES, SIGNIFICANTLY	YES, SOMEWHAT	NO, IT DOESNOT MATTER	NO, I DONT TRUST IT	
MALE	14	25	17	4	60
FEMALE	18	46	14	13	91
	32	71	31	17	151

EXPECTED OUTCOME

	YES, SIGNIFICANTLY	YES, SOMEWHAT	NO, IT DOESNOT MATTER	NO, I DONT TRUST IT
MALE	12.71523179	28.21192053	12.31788079	6.754966887
FEMALE	19.28476821	42.78807947	18.68211921	10.24503311

(O-E)²/E

	YES, SIGNIFICANTLY	YES, SOMEWHAT	NO, IT DOESNOT MATTER	NO, I DONT TRUST IT
MALE	0.1298151214	0.3656763983	1.779708752	1.123594338
FEMALE	0.08559238774	0.2411053176	1.173434342	0.7408314319

X2	5.639758089
DF	1
P-VALUE	0.01755762951
CHI- SQUARE	0.9999600274

INTERPRETATION:

The Chi-Square test result ($p = 0.0175$) indicates that the p-value is less than 0.05. Therefore, the null hypothesis is rejected, and it can be concluded that there is a statistically significant association/difference between the variables.

4.27. ANOVA

4.27.1: SHOWING ANOVA FOR FREQUENCY AND AI IN CUSTOMER SUPPORT AND SERVICE QUALITY

NULL HYPOTHESIS (H_0):

There is no significant association between How frequently customers shop online and AI in Customer Support and Service Quality. (Both variables are independent.)

ALTERNATIVE HYPOTHESIS (H_1):

There is a significant association between How frequently customers shop online and AI in Customer Support and Service Quality. (Both variables are dependent.)

1)ANOVA: SINGLE FACTOR

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	64.15453	5	12.83091	9.846185	3.53E09	2.22405
Within Groups	1172.821	900	1.303135			
Total	1236.976	905				

INTERPRETATION:

Since the P-value is far less than 0.05 and the calculated F-value (9.846) is greater than the critical F-value (2.224), we reject the null hypothesis.

4.27.2: SHOWING ANOVA FOR FREQUENCY AND AI-DRIVEN PERSONALIZATION AND CUSTOMER EXPERIENCE

NULL HYPOTHESIS (H_0):

There is no significant difference in customer experience with AI-driven personalization across different shopping frequency groups.

ALTERNATIVE HYPOTHESIS (H_1):

There is a significant difference in customer experience with AI-driven personalization across different shopping frequency groups.

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	71.21413	5	14.24283	10.22925	1.5E-09	2.22405
Within Groups	1253.126	900	1.392362			
Total	1324.34	905				

INTERPRETATION:

Since the calculated F (10.229) is greater than the critical F (2.224) and the p-value is much smaller than 0.05, we reject the null hypothesis.

4.27.3: SHOWING ANOVA FOR FREQUENCY AND AI IN CUSTOMER ENGAGEMENT ON E-COMMERCE PLATFORMS

NULL HYPOTHESIS (H_0):

There is no significant difference in customer engagement through AI across different shopping frequency groups.

ALTERNATIVE HYPOTHESIS (H_1):

There is a significant difference in customer engagement through AI across different shopping frequency groups.

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	58.5883	5	11.71766	8.24077	1.27E-07	2.22405
Within Groups	1279.722	900	1.421913			
Total	1338.31	905				

INTERPRETATION:

Since the calculated F-value (8.24) is greater than the critical F-value (2.22) and the p-value is far less than 0.05, we reject the null hypothesis.

4.27.4: SHOWING ANOVA FOR FREQUENCY AND AI-BASED SAFETY AND SECURITY FEATURES ON CUSTOMER TRUST AND SATISFACTION

NULL HYPOTHESIS (H_0):

There is no significant difference in customer trust and satisfaction with AI-based safety and security features across different shopping frequency groups.

ALTERNATIVE HYPOTHESIS (H_1):

There is a significant difference in customer trust and satisfaction with AI-based safety and security features across different shopping frequency groups.

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	51.64238	5	10.32848	7.722661	4.01E-07	2.22405
Within Groups	1203.682	900	1.337425			
Total	1255.325	905				

INTERPRETATION:

Since the calculated F-value (7.72) is greater than the critical F-value (2.22) and the p-value is far less than 0.05, we reject the null hypothesis.

CHAPTER 05

FINDINGS

- The majority of respondents (68.9%) are aged between 18 and 25
- The majority of respondents are female (60.3%)
- The majority of the respondents (49%) are undergraduates
- The majority of respondents (60.9%) are students
- Most respondents (49%) shop online occasionally
- The majority of respondents (64.2%) belong to families with 4 members
- Most respondents earn either below ₹20,000 (32.5%) or above ₹50,000 (31.8%) monthly
- Amazon is the most preferred e-commerce platform among respondents (51.7%)
- Clothing and fashion products are the most purchased online by respondents (52.3%)
- Cash on Delivery (49%) is the most preferred payment method among respondents
- Most respondents (36.4%) have been using online shopping for 4–6 years
- A majority of respondents (44.4%) sometimes use AI-powered customer support while shopping online
- Most respondents (62.9%) sometimes trust AI recommendations for product suggestions
- Most respondents (38.4%) rarely use AI-powered voice or image search for product discovery
- Most respondents (39.7%) remain neither agree nor disagree about the importance of a personalized shopping experience
- A majority of respondents (56.3%) believe AI-powered recommendations sometimes help them make better purchase decisions
- Most respondents (37.1%) feel AI-based customer support is somewhat better than human support
- The majority of respondents (64.9%) are somewhat concerned about AI-based security and privacy in online shopping
- Most respondents (47%) somewhat trust online payments due to AI-driven fraud detection systems
- Most respondents have limited exposure to AI-driven security features, with 31.8% experiencing them rarely
- Most respondents (39.1%) would probably recommend AI-driven e-commerce platforms
- Most respondents (33.1%) agree that AI-powered chatbots are effective and fast in handling customer queries
- Most respondents (38.4%) agree that AI-based virtual assistants improve the efficiency of

customer service

- Most respondents (35.8%) are neither agree nor disagree about AI reducing customer waiting time
- Most respondents (36.4%) agree that AI-driven customer support enhances problem resolution accuracy
- Most respondents (42.4%) are neither agree nor disagree about how AI improves their overall experience with e-commerce customer support
- Most respondents (34.4%) are neither agree nor disagree about the relevance and usefulness of AI-powered product recommendations
- Most respondents (35.1%) are neither agree nor disagree about the impact of AI-based targeted promotions and discounts on purchasing behavior
- Most respondents (32.5%) are neither agree nor disagree about AI customizing their shopping journey based on browsing and purchase history
- Most respondents (37.7%) agree that AI helps them discover new products matching their preferences
- Most respondents (36.4%) agree that AI-driven personalization improves their shopping experience
- Most respondents (37.1%) are neither agree nor disagree about AI-powered chatbots making their shopping experience more interactive
- Most respondents (33.1%) are neither agree nor disagree about AI-driven personalized notifications keeping them engaged
- Most respondents (33.1%) agree that AI-powered voice and visual search make it easier to find products
- Most respondents (40.4%) agree that AI enhances their engagement by offering customized deals and recommendations
- Most respondents (33.1%) are neither agree nor disagree about AI-driven marketing strategies keeping them connected to e-commerce platforms
- Most respondents (39.7%) are neither agree nor disagree about AI's role in detecting fraudulent activities on e-commerce platforms
- Most respondents (35.1%) are neither agree nor disagree about AI-based security measures increasing their trust in online shopping
- Most respondents (39.7%) are neither agree nor disagree about AI preventing unauthorized access to their e-commerce accounts
- Most respondents (39.1%) are neither agree nor disagree about AI-powered identity verification enhancing online transaction security

- Most respondents (35.1%) are neither agree nor disagree about AI-driven security features giving them confidence in making online purchases
- The Chi-Square test shows no significant association between gender and the usage of AI-powered customer support while online shopping.
- There is a statistically significant association between gender and trust in online payments due to AI-driven fraud detection systems.
- There is a significant association between shopping frequency and AI in customer support and service quality, meaning customers' perceptions vary based on how often they shop online.
- There is a significant difference in customer experience with AI-driven personalization across shopping frequency groups, showing that personalization impacts customers differently based on how often they shop online.
- There is a significant difference in customer engagement through AI across shopping frequency groups, indicating that engagement levels vary depending on how often customers shop online.
- There is a significant difference in customer trust and satisfaction with AI-based safety and security features across shopping frequency groups, meaning perceptions of security vary depending on how often customers shop online.

CHAPTER 06

SUGGESTIONS

COMPANIES:

- Companies can use AI to analyze large volumes of data to understand customer preferences better.
- Companies can implement AI chatbots to reduce customer service costs and improve response efficiency.
- Companies can adopt AI-powered fraud detection systems to secure transactions.

CUSTOMERS:

- Customers get personalized recommendations that save time and improve shopping experience.
- Customers benefit from 24/7 instant support without waiting for human agents.
- Customers enjoy consistent product availability and faster delivery.

CHAPTER 07

CONCLUSION

This study aims to analyze the role of Artificial Intelligence in enhancing customer experience on e-commerce platforms, including how AI tools such as personalized recommendations, chatbots, fraud detection, faster deliveries, and virtual try-on features improve convenience and trust. Based on today's conditions, where customers expect speed, security, and personalized service, the study concludes that AI plays a crucial role in making online shopping more efficient, reliable, and satisfying. It not only improves customer satisfaction but also helps e-commerce businesses remain competitive in the rapidly growing digital marketplace.

BIBLIOGRAPHY

Chen, & Zhang, Artificial intelligence in e-commerce: “Applications and implications for customer experience”, (2021), Vol. 61, ISSN 0969-6989.

Davenport, Guha, Grewal. How artificial intelligence will change the future of marketing, (2020), Vol. 48, Issue 1, ISSN 0092-0703.

Ghosh, “AI-driven personalization in online retailing”, (2022), Vol. 12, Issue 4, ISSN 2347-4696.

Huang, M., Rust, R. T., & Maksimovic, V. The feeling economy: Managing in the next generation of artificial intelligence (AI), (2019), Vol. 61, Issue 4, ISSN 0008-1256.

Jain, R., & Singh, S. Artificial intelligence and customer relationship management in e-commerce, (2020), Vol. 21, Issue 6, ISSN 0972-1509.

Kapoor, R., & Dwivedi, Y. K. AI-enabled chatbots and their role in shaping customer experience, (2022), Vol. 142, ISSN 0148-2963.

Kumar, V., Dixit, A., Javalgi, R. G., & Dass, M, Digital transformation of customer experience with AI in retail, (2021), Vol. 32, Issue 5, ISSN 1757-5818.

Li, X., & Sun, Y. The role of artificial intelligence in improving e-commerce security and fraud detection, (2020), Vol. 22, Issue 5, ISSN 1387-3326.

Mishra, P., & Sharma, K. AI and augmented reality in online shopping: A customer-centric approach, (2021), Vol. 49, Issue 9, ISSN 0959-0552.

Nguyen, T., & Simkin, L. The dark side of artificial intelligence in retailing, (2020), Vol. 54, ISSN 0969-6989.

Patel, D., & Mehta, R. AI-powered recommendation systems in e-commerce platforms, (2022), Vol. 13, Issue 3, ISSN 0976-2191.

Prasad, S., & Srivastava, S. Customer trust in AI-enabled e-commerce services, (2021), Vol. 36, Issue 13, ISSN 0885-8624.

Singh, A., & Choudhury, P. Machine learning applications for improving delivery and logistics in e-commerce, (2020), Vol. 9, Issue 6, ISSN 2050-7399.

Wang, Y., & Yu, C. Artificial intelligence adoption in e-commerce: Opportunities and challenges for customer engagement, (2022), Vol. 17, Issue 2, ISSN 0718-1876.

Zhang, T., & Luo, J., AI-based personalization and its impact on customer satisfaction in online retail, (2021), Vol. 120, ISSN 0747-5632.

WEBLIOGRAPHY

1. <https://www.hashstudioz.com/blog/ai-in-customer-experience-a-game-changer-for-e-commerce>
2. <https://medium.com/@vaishnaviyada/enhancing-customer-experiences-with-ai-in-e-commerce-c0cfe6868d13>
3. <https://www.247commerce.co.uk/ecommerce-insights/industry-insights/how-ai-is-enhancing-e-commerce-user-experiences>
4. <https://sapient.pro/blog/ways-ai-is-transforming-ecommerce>
5. <https://esoftskills.com/ai-in-e-commerce-the-ultimate-guide>
6. <https://hutechsolutions.com/blogs/ai/ai-in-ecommerce-customer-experience>
7. <https://ecommercefastlane.com/the-rise-of-ai-in-e-commerce>
8. <https://www.shipbuddies.com/how-ai-is-enhancing-the-customer-experience-in-ecommerce>
9. <https://fiftyminds.com/the-role-of-ai-in-e-commerce-enhancing-the-customer-experience>
10. <https://newo.ai/insights/ai-in-e-commerce-use-cases>

ANNEXURE

1. Age Group:

- a) Below 18
- b) 18-25
- c) 26-35
- d) 36-45
- e) Above 40

2. Gender:

- a) Male
- b) Female
- c) Prefer not to say

3. Educational Qualification:

- a) High School or below
- b) Undergraduate
- c) Postgraduate
- d) Other

4. Occupation:

- a) Student
- b) Employed
- c) Self-employed
- d) Unemployed

5. How frequently do you shop online?

- a) Daily
- b) Weekly
- c) Monthly
- d) Occasionally

6. Number of members in family

- a) 1
- b) 2
- c) 3
- d) 4
- e) more than 4

7. Average monthly income

- a) below 20,000
- b) 21000-30000
- c) 31000-40000
- d) 41000-50000
- e) above 50000

8. Which e-commerce platforms do you use the most?

- a) Amazon
- b) FLIPKART
- c) Meesho
- d) Others (please specify)

9. What type of products do you buy most in online?

- a) Electronics
- b) Clothing & Fashion
- c) Groceries & Essentials
- d) Books & Stationery
- e) Other

10. Which payment method do you prefer for online shopping?

- a) Credit/Debit Card
- b) Digital Wallet (PayPal, G- Pay, etc.)
- c) Cash on Delivery
- d) Bank Transfer

11. How long have you been shopping online?

- a) Less than a year
- b) 1-3 years
- c) 4-6 years
- d) More than 6 years

12. Do you use AI-powered customer support (e.g., chatbots) when shopping online?

- a) Always
- b) Often
- c) Sometimes
- d) Never

13. Do you trust AI recommendations for product suggestions?

- a) Always

b) Often

c) Sometimes

d) Never

14. Have you used AI-powered voice or image search for product searches?

a) Yes, frequently

b) Yes, occasionally

c) Rarely

d) Never

15. How important is personalized shopping experience for you?

a) Very important

b) Somewhat important

c) Neutral

d) Not important

16. Do AI-powered recommendations help you make better purchase decisions?

a) Yes, always

b) Yes, sometimes

c) No, rarely

d) No, never

17. Do you feel AI-based customer support is better than human customer support?

a) Yes, much better

b) Yes, somewhat better

c) No difference

d) No, human support is better

18. How concerned are you about AI-based security and privacy in online shopping?

a) Very concerned

b) Somewhat concerned

c) Neutral

d) Not concerned

19. Do AI-driven fraud detection systems increase your trust in online payments?

a) Yes, significantly

b) Yes, somewhat

c) No, it does not matter

d) No, I don't trust it

20. Have you ever experienced an AI-driven security feature (e.g., automated fraud detection) in online shopping?

a) Yes, frequently

b) Yes, occasionally

- c) Rarely
- d) Never

21. Would you recommend AI-driven e-commerce platforms to others based on your experience?

- a) Yes, definitely
- b) Yes, probably
- c) Not sure
- d) No

22.

AI in Customer Support and Service Quality	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI-powered chatbots provide quick and effective responses to customer queries.					
AI-based virtual assistants improve the efficiency of customer service.					
AI reduces customer waiting time by automating responses and solutions.					
AI-driven customer support enhances problem resolution accuracy.					
AI improves my overall experience with e-commerce customer support services.					

23.

AI-Driven Personalization and Customer Experience	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI-powered product recommendations are relevant and useful for me.					
AI-based targeted promotions and discounts encourage me to purchase more.					
AI customizes my shopping journey based on my browsing and purchase history.					
AI helps me discover new products that match my preferences.					
AI-driven personalization improves my shopping experience.					

24.

AI in Customer Engagement on E-commerce Platforms	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI-powered chatbots make my shopping experience more interactive.					
AI-driven personalized notifications keep me engaged with the platform.					
AI-powered voice and visual search make it easier to find products.					
AI enhances my engagement by offering customized deals and recommendations.					

AI-driven marketing strategies (emails, messages, etc.) keep me connected to e-commerce platforms.					
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25..

AI-Based Safety and Security Features on Customer Trust and Satisfaction	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI helps in detecting fraudulent activities on e-commerce platforms.					
AI-based security measures increase my trust in online shopping.					
AI prevents unauthorized access to my e-commerce accounts.					
AI-powered identity verification enhances the security of online transactions.					
AI-driven security features give me confidence in making online purchases.					